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Global firms and emissions: Investigating the dual channels of emissions abatement☆

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ABSTRACT

This paper finds that both importing and exporting reduce firm-level emission intensities. We develop a theoretical model in which firms jointly determine the extent of abatement investment as well as whether or not to import intermediate inputs and export final goods. The model demonstrates a complementarity between trading decisions and abatement investment such that trading firms invest more in abatement technology and feature lower emission intensities. Using Chinese firm-level data, we estimate a theory-based regression model with instrument variables to account for the endogeneity issue associated with firms' trading statuses. Our regression results show that both importing and exporting reduce firm-level emission intensities by over 15 percent.

1. Introduction

How does international trade impact firms' environmental performance? In this paper, we answer this question with a model that links firms' import, export, and abatement decisions in a single framework. Our theory suggests that firms have a greater incentive to invest in abatement technology when they become importers, exporters, or both, and hence their firm-level emission intensities, defined as mass of emissions normalized by the real value of output, decline in response to trade liberalization. Our empirical results confirm that both importing and exporting contributed to significant reduction of firm-level emission intensities in China in the first decade of the 21st century.

Although voluminous studies have shown that *exporting* improves firm-level environmental outcomes (e.g., Holladay (2016), Forslid et al. (2018), Shapiro and Walker (2018), Rodrigue et al. (2022a)), there is scant evidence on whether *importing* also generates a similar effect.[2] Against this backdrop, our paper aims to instate the key role that importing plays in reducing firms' emission intensities. Our contributions in this paper can be summarized as follows. First, we propose a theoretical model that elucidates how

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[2] One of the few papers that investigate the impact of importing on firm-level emissions is Cherniwchan (2017), who studies the effect of NAFTA on US firms' emissions. The key advantage of our paper is that we observe each firm's importer status and imported products. We posit that importing is a key aspect of trade liberalization, since numerous studies have shown that importing has a strong implication on firms' non-environmental outcomes, such as productivity (Amiti and Konings, 2007), R&D decisions (Bøler et al., 2015) and currency invoicing (Yang and Kwon, 2021).

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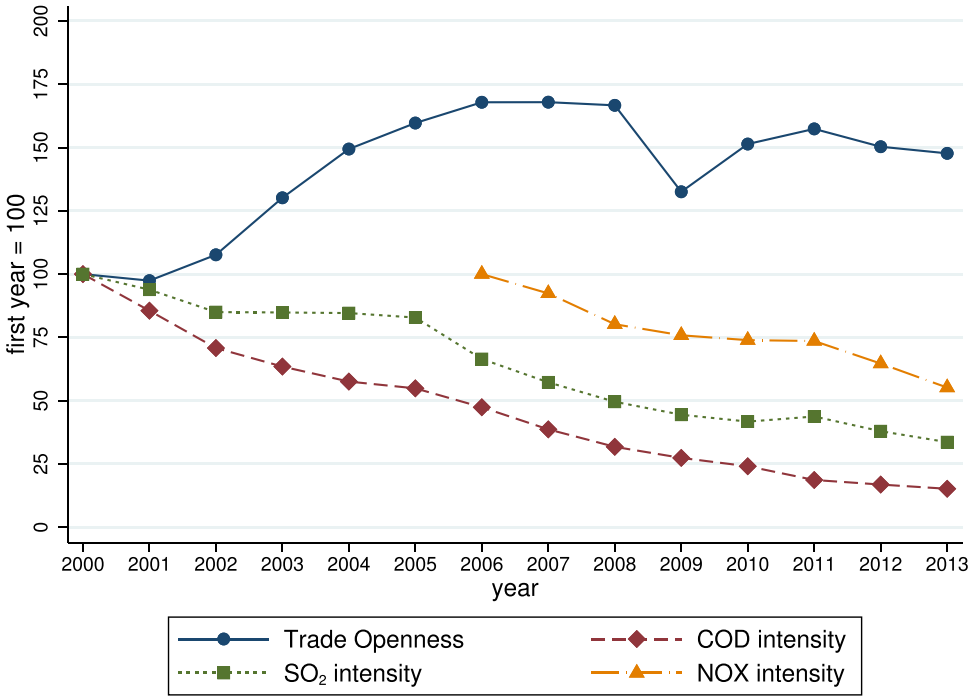

Fig. 1. Trends of trade openness and emission intensities in China since 2000.
Notes:
(i) This figure shows the trends in trade openness and the three major types of emission intensities that are the focus of our paper. Trade openness is measured by the sum of nominal imports and exports divided by the nominal GDP, and emission intensities are measured by the weights of respective substances divided by the real GDP. All series are normalized by the respective levels in the first year in which the data are available.
(ii) To reduce the aggregation errors due to slight differences in firms' reporting criteria across years, for each pair of adjacent years, we calculate the percentage change of total emissions by firms that appear in both years of our data. We compound the year-by-year percentage changes to generate the emission data used in the figure.
Source: Chinese Statistical Yearbook and the CESD (China's Environmental Statistical Database) dataset.

firms' decision to import intermediate inputs leads to an increase in abatement investment and, hence, a reduction in their emission intensities. An extension of our model that allows firms to export final goods shows that both importing and exporting can reduce firms' emission intensities. Intuitively, if firms pay certain fees for emissions, firms that have become larger due to importing or exporting (or both) have greater incentive to invest in abatement to lower emissions per unit of output.

For our second contribution, we exploit the structural decomposition of firm-level emission intensities guided by our model and estimate the relevant parameters with Chinese firm-level data. Our IV regression results show that both importing and exporting led to sizable reduction in firm-level emission intensities across three different types of emissions: chemical oxygen demand (COD), sulfur dioxide (SO_2), and nitrogen oxide (NOX). Furthermore, we validate our theoretical framework by confirming its mechanism with data. We show that both importing and exporting significantly increase Chinese firms' water treatment capacities and their removal of COD, SO_2 and NOX emissions.

Lastly, we adopt a novel empirical strategy to demonstrate that the import-side mechanism is not driven by pollution offshoring hypothesis (POH) — a competing mechanism that is known to create a negative link between importing and emission intensities. Intuitively, we show that importing reduces firms' emission intensities even if we limit our attention to a subset of imported intermediate inputs that do not cause pollution offshoring.[3] These results further substantiate our theoretical framework which suggests that firms' emission abatement is reinforced by international trade.

Overall, our paper demonstrates that both importing and exporting – dual characteristics of global firms – contribute to lower emission intensities at the firm level. We believe that our study on Chinese firms during the first decade of the twenty-first century is particularly relevant for other developing countries undergoing similar trends of trade liberalization.[4]

In Fig. 1, we present the trends in trade openness and three major types of emission intensities in our study. We observe that China's trade openness (sum of imports and exports normalized by the nominal GDP) increased significantly between 2000 and

[3] See Cherniwchan et al. (2017) for discussions on the pollution offshoring hypothesis. To our knowledge, Cherniwchan (2017) provides the first empirical evidence of the pollution offshoring hypothesis.

[4] China experienced an episode of remarkable trade liberalization and the subsequent rise in environmental emissions raised concerns among scholars and the Chinese government. See Pierce and Schott (2016), Nordhaus (2015) and Shi and Xu (2018) for relevant discussions.

Table 1
Differences in emission intensities by trading type (Average firms in 2010).

	Δ log (emission intensity)		
	COD	SO_2	*NOX*
Domestic	0.00	0.00	0.00
Import-only	−0.77	−0.98	−1.01
Export-only	−0.97	−1.25	−1.03
Both	−1.49	−2.14	−1.92

Notes:
(i) The table shows the log-differences of emission intensities of average firms in each trading type relative to domestic firms in 2010. The emission intensity is defined as the emissions (measured in tonnes) normalized by the annual output. The log-difference is computed relative to the "domestic" group. The similar tables for other years, available on request, remain qualitatively the same.
(ii) "Domestic" firms are those that do not import or export in a particular year; "import-only" firms are those that engage in import transactions without any export transactions; "export-only" firms are those that engage in export transactions without any import transactions, and the category "both" comprises firms that engage in both import and export transactions.
(iii) Data sources: Emission data are from the CESD dataset. Output data are from the ASIE (Annual Survey of Industrial Enterprises) dataset. Importer and exporter statuses are from the Chinese customs dataset.

2013. In contrast, the levels of emission intensities (total mass of emissions normalized by the real GDP) fell sharply during the same period.[5,6]

For a micro-level comparison, we present in Table 1 the log-differences of firms' average emission intensities in each trading type relative to domestic firms. We consider four types of firms: domestic, importing-only, exporting only, and firms that both import and export. Note that both importing and exporting firms exhibit considerably lower emission intensities than domestic firms. Existing literature (e.g., Holladay (2016), Forslid et al. (2018), Shapiro and Walker (2018), and Rodrigue et al. (2022a)) documents that *exporting* firms exhibit lower emission intensities than domestic firms, but less is well established as to how and why *importing* also reduces firm level emission intensities.[7] Moreover, we observe that firms that both import and export exhibit the *lowest* pollution intensities among all types of firms, suggesting that the impacts from importing and exporting can be "stacked" together.

Motivated by Table 1, we investigate whether international trade has a causal impact of lowering firm-level emission intensities. We propose a novel framework in which firms' import, export and abatement decisions are jointly determined. The model predicts that firms engaging in import and export invest more in abatement and thus feature lower emission intensities than they would otherwise do if they were not trading. Capitalizing on detailed Chinese firm-level emissions data, we then use a set of IV regressions to show that the model prediction is borne out in the data and that the results are robust against the competing POH hypothesis. Furthermore, we empirically assess the potential bias in the estimates of exporter coefficients if researchers fail to include both importer and exporter statuses in the same regression.

Related Literature. Our paper is closely related to Rodrigue et al. (2022a) which show that enhanced exporting opportunities since the accession to the WTO reduced firm-level emissions in China by over 36 percent. Our paper complements Rodrigue et al. (2022a) by emphasizing the role of importing and provides a comprehensive understanding of how international trade affects firm-level emission intensities. Crucially, we show that importing has negative effects on emission intensities in addition to the export channel.

Our paper contributes to the emerging literature that explores the impact of international trade on firm-level emissions.[8] Holladay (2016) and Cui et al. (2016) show that US exporting firms exhibit lower emission intensities after controlling for individual firm characteristics. A key distinction of our paper is to adopt IV to establish the causal effects on emission intensities.[9] Forslid et al. (2018) endogenize firm-level investment in abatement technology in a Chaney–Melitz style heterogeneous firm model.[10] They show that exporting firms invest additionally in abatement technology, and thus they exhibit lower emission intensities than otherwise comparable domestic firms. Shapiro and Walker (2018) develop a general equilibrium model of endogenous trade and emission abatement. Armed with the estimated parameters, Shapiro and Walker (2018) persuasively show that the declining emissions in the

[5] Data on NOX are not available before 2006. However, we cautiously conjecture that the NOX intensity also experienced a comparable decline since 2000 as other emission intensities.

[6] Fan et al. (2019) perform the scale-composition-technique decomposition à la Levinson (2015) and show that the decline in COD emissions is mainly attributed to the technique channel. In our main analysis, we always include firm, industry-year and province-year fixed effects so that the impact of international trade is identified from within-firm variation after accounting for shocks common to comparable firms. This approach allows us to fully account for the shift in the industrial structure over time.

[7] Cherniwchan (2017) is one of few papers that empirically examine importing and reduced emission intensities at the firm level. Cherniwchan et al. (2017) propose a formal theory of pollution offshoring hypothesis that produces a link between importing and reduced emission intensities.

[8] We refer readers to Cherniwchan et al. (2017) for a survey on international trade, emissions and the environment.

[9] Rodrigue et al. (2022a), which also adopts an IV approach, albeit a different one, is a key exception to this statement.

[10] Specifically, we refer to the innovative modification by Chaney (2008) of Melitz (2003)-type heterogeneous firm model by assuming that the distribution of innate firm productivity follows a Pareto distribution.

US since the 1990s is attributable to more stringent environmental protection policies. Our paper is also related to Cherniwchan (2017), which finds that the emissions of PM_{10} and SO_2 from the US plants dropped significantly as a result of NAFTA. In comparison, we theoretically demonstrate that importing enhances firms' abatement investment and confirm this mechanism empirically.[11]

More broadly, there has been sustained research interest to investigate the interaction between globalization and the environment.[12] Seminal papers by Copeland and Taylor (1994) and Grossman and Krueger (1995) were followed up by subsequent empirical findings including (but not limited to) Antweiler et al. (2001), Levinson (2009), Bombardini and Li (2020) and Najjar and Cherniwchan (2021). Overall, the literature does not provide a consensus on how globalization affects the environment. As Cherniwchan and Taylor (2022) suggest, there are numerous margins of trade that can affect the environment, and thus studies should focus on specific margins. Our paper uses rich firm-level information to investigate the import margin of trade, which is equally important as the export margin well established in the literature.

The remainder of the paper is organized as follows. In Section 2, we present our baseline model to illustrate why firms endogenously reduce emission intensities if they import intermediate inputs or export final goods. We describe our datasets in Section 3. Our empirical strategies and results are presented in Section 4. Section 5 concludes.

2. A model of international trade and abatement investment

In this section, we describe a firm's problem involving international trade and emission abatement investment. We characterize the firm's optimal level of abatement investment and discuss how it relates to import and export decisions. For expositional clarity, we first discuss the interaction between *import* and abatement investment decisions. Then, we introduce *export* decisions to construct our full model. The key prediction of our model is that import and export decisions amplify firms' abatement investment.

2.1. Demand and utility

Consider a model of two countries — home and foreign. In the home country, the representative consumer has a quasi-linear utility given by

$$U = q_0 + E\log(Q), \tag{1}$$

where q_0 is the numéraire good, the price of which is normalized to 1, and E (exogenous in our model) is the expenditure on differentiated good Q. We adopt a quasi-linear utility function so that the numéraire good's consumption fully absorbs the economy-wide income effects, which are not central to our analysis.

We assume that the numéraire good is produced with $1/w$ units of labor under constant returns to scale (CRS) and is sold in a competitive market. We also assume that the labor endowment in the home country is sufficient to ensure a positive level of production for q_0. These two assumptions pin down the wage at w, which is treated as a constant in the remainder of our analysis.

Let I denote the income of the representative consumer in the home country.[13] Utility maximization subject to the constraint $q_0 + PQ \leq I$ leads to $E = PQ$ where P is the ideal price index for consumption Q.[14] We further express Q as the CES aggregate of differentiated goods

$$Q = \left[\int q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}},$$

where $q(\omega)$ is the consumption amount for variety ω, and σ ($\sigma > 1$) is the elasticity of substitution across varieties of goods. The solution to the representative consumer's variety-wise utility maximization problem leads to the following expression of the ideal CES price index

$$P = \left[\int p(\omega)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}},$$

where $p(\omega)$ is the price of variety ω. Using the expression of the price index, we can derive the variety-specific demand curve as $q(\omega) = EP^{\sigma-1}p(\omega)^{-\sigma}$.

In the foreign country, the utility of a representative consumer is also given by Eq. (1) with respective foreign variables. We assume further that the numéraire good is produced with the same technology in the foreign country and that the numéraire good is traded costlessly between the home and foreign countries. This also pins down the foreign wage at w. The variety-wise demand in the foreign country can be expressed similarly as $q^*(\omega) = E^*P^{*\sigma-1}p^*(\omega)^{-\sigma}$.

[11] Other studies investigating the impact of globalization on developing countries show that (1) emission content of exports aggravates the emissions and mortality level at the prefecture level in China (Bombardini and Li, 2020); (2) more stringent environmental protection regulations in China discourage exports (Shi and Xu, 2018) and labor demand (Liu et al., 2017); (3) import tariff liberalization has been in favor of emission-intensive industries and thus shifting production abroad (Shapiro, 2021); and (4) China's eleventh five-year plan was effective in reducing COD emissions (Fan et al., 2019).

[12] See Copeland and Taylor (2004), Copeland et al. (2022) and Cherniwchan and Taylor (2022) for a comprehensive review of the subject.

[13] The home country's income comprises labor income and producer surplus. In our model, the numéraire good in the quasi-linear utility function absorbs the entire income effects as long as $I > E$. Since we treat the total expenditure on the differentiated good as a constant, the endogenous determination of I is no longer central to our analysis.

[14] We also rely on the prior assumption that there is a sufficient amount of labor to prevent a corner solution where $q_0 = 0$.

2.2. Production

Each firm in our model is a monopoly of a single variety of good ω. An importing firm's production function can be expressed as

$$q = \phi\,[(1-\theta)l]^{\upsilon_l}\left(\upsilon_d m_d^{\frac{\eta-1}{\eta}} + \upsilon_f m_f^{\frac{\eta-1}{\eta}}\right)^{\frac{\eta\upsilon_m}{\eta-1}}, \tag{2}$$

where ϕ is the Hicks-neutral technology, θ is the share of labor dedicated to emission reduction, l is the labor input, m_d is the domestic intermediate input, m_f is the foreign intermediate input, υ_i is the weight on each input i that satisfies $\upsilon_l + \upsilon_m = 1$, and $\eta > 1$ is the elasticity of substitution between domestic and foreign inputs.[15] We assume that domestic and foreign intermediate inputs are produced under a CRS technology with c_d/w and c_f/w units of labor, respectively, and are sold in a competitive market. This suggests that the unit costs of m_d and m_f are given by c_d and c_f, respectively.

For clarity, we refer to a firm that does not import intermediate inputs or export final goods as a *domestic* firm. For a domestic firm, the production function in Eq. (2) reduces to

$$q = \phi\,[(1-\theta)l]^{\upsilon_l}\,\upsilon_d^{\frac{\eta\upsilon_m}{\eta-1}}\,m_d^{\upsilon_m}, \tag{3}$$

which is a special case of production function (2) when $m_f = 0$.[16]

Let z denote the mass of emissions subject to a constant marginal pollution tax τ. Following Forslid et al. (2018), we assume that z is generated with the following technology:

$$z = \frac{(1-\theta)^{\frac{1}{\alpha}}}{h(f_A)}\,l, \tag{4}$$

where α $(0 < \alpha < 1)$ is a scale parameter used to prevent corner solutions, and $h(f_A)$ is the emission abatement discount factor that is increasing in the firm's abatement investment f_A.[17] Eq. (4) shows that, holding constant θ and f_A, the level of emissions z grows linearly with the firm's labor input.[18] The firm can reduce emissions by increasing the share of labor dedicated to emission reduction (higher θ) and increasing an up-front abatement investment f_A.[19]

Eq. (4) demonstrates the IRS property of the abatement investment, which is crucial for our following analysis. Since total emission tax is given by $\frac{(1-\theta)^{\frac{1}{\alpha}}}{h(f_A)}l\tau$, $h(f_A)$ is akin to a discount factor that lowers the marginal tax of emissions by a constant factor of $h(f_A)$ for any level of total emissions.[20,21] Thus, it is profitable for larger firms with higher production levels to make a greater investment in f_A to reduce emission tax per unit of production. In contrast, smaller firms choose to make less abatement investment f_A leading to their higher emission tax per unit of production.

Combining Eqs. (2) and (4) at the *optimal* level of θ, we derive an alternative expression for the production function:

$$q = \phi h(f_A)^{\alpha\upsilon_l}\,l^{(1-\alpha)\upsilon_l}\,z^{\alpha\upsilon_l}\left(\upsilon_d m_d^{\frac{\eta-1}{\eta}} + \upsilon_f m_f^{\frac{\eta-1}{\eta}}\right)^{\frac{\eta\upsilon_m}{\eta-1}}. \tag{5}$$

A direct interpretation of this expression is that the firm's production process can be thought of as choosing the optimal levels of variable inputs l, z, m_d and, m_f and the abatement investment f_A to produce the amount q of final goods.

Consistent with Forslid et al. (2018) and Shapiro and Walker (2018), Eq. (5) permits dual interpretations of the role of emissions in production. Specifically, we can regard emissions as a byproduct of production through the lens of Eq. (4). Another way to interpret the source of emissions, by Eq. (5), is that emissions are a variable input, the cost of which is given by the marginal

[15] The gross substitutability assumption between domestic and foreign inputs is widely adopted in the literature (e.g., Bøler et al. (2015)). This can be micro-founded by assuming that both foreign and domestic input prices follow independent probabilistic distribution, and the firm aims to source from the lowest-cost provider à la Antras et al. (2017). By seeking the input providers from multiple regions, the firm can lower the expected input cost.

[16] We remind readers that υ_d and υ_f are free parameters to adjust the relative weights of domestic and foreign inputs, and have no roles in our model. In particular, if we set both υ_d and υ_f to be 1, none of our results will change.

[17] We impose the parameter range $\alpha \in (0,1)$ to ensure that the firm's problem leads to an internal solution. The cost minimization problem for θ leads to its optimal level given by

$$\theta = 1 - \left[w\frac{\alpha}{1-\alpha}\frac{h(f_A)}{\tau}\right]^{\alpha}.$$

Clearly, the right-hand side is not well defined if $\alpha < 0$ or $\alpha \geq 1$. If $\alpha = 0$, the optimal solution for θ is a corner one $(\theta = 1)$.

[18] Whereas labor input per se does not create emissions in reality, one can interpret labor in our model as a composite of inputs that firms use in general.

[19] More generally, we posit that emissions z results from the firm's production processes. In our model, this translates into z increasing linearly with labor, which is the only source of value-added.

[20] Conceptually, there may be two real-world interpretations regarding the function $h(f_A)$. One interpretation is that the firm installs treatment facilities to neutralize the final emissions. The other interpretation is that the firm streamlines the production process to lower emissions to begin with. We argue that both interpretations are consistent with our model, and thus we do not take a particular stance. When we test the model mechanism in the empirical section, however, we are only able to test the first channel.

[21] In our model, we simply model stringency of environmental policy as the level of marginal emission tax τ. In reality, there can be other forms of regulations such as emissions cap or cap-and-trade. Kwon et al. (2021b) show that other environmental emission regulations can be interpreted as the government determining the shadow cost of emissions, which plays the same role as the emissions tax in our model.

emission tax τ. Thus, our baseline model adopts a flexible production-side specification and does not take a specific stance on whether emissions are a byproduct of production or they explicitly become an input for production.

Similarly for a domestic firm, we can express the alternative production function as

$$q = \phi h(f_A)^{\alpha v_l} l^{(1-\alpha)v_l} z^{\alpha v_l} v_d^{\frac{\eta v_m}{\eta-1}} m_d^{v_m}, \tag{6}$$

which is a special case of Eq. (5) when m_f is reduced to zero. Having specified the firm's demand function and the production function, we characterize the firm's optimal import and abatement decisions in the next section.

2.3. The firm's import and abatement decisions

Let $\pi^D(\phi)$ and $\pi^M(\phi)$ denote, respectively, the operating profit of a domestic firm and that of an importing firm with a productivity level given by ϕ. Operating profit is defined as the firm's revenue net of its variable costs (labor cost, intermediate input cost and pollution tax) and the fixed costs of abatement and importing:

$$\pi^D(\phi) = pq - wl - \tau z - c_d m_d - f_A^D(\phi),$$

$$\pi^M(\phi) = pq - wl - \tau z - c_d m_d - c_f m_f - f_A^M(\phi) - f_I,$$

where p is the output price, $f_A^D(\phi)$ is the fixed cost of abatement investment for a domestic firm with a productivity level ϕ, $f_A^M(\phi)$ is the fixed cost of abatement investment for an importing firm, and f_I is the fixed cost of importing.[22]

As we assume that production features the CRS property (see Eqs. (5) and (6)), we can solve the firm's profit maximization problem in two steps. In the first step, we solve for the firm's optimal input bundle to produce a unit of output. This leads to an intuitive expression for the marginal cost independent of the firm's scale. In the second step, we solve for the firm's optimal level of production q and its optimal level of abatement investment f_A.

To prevent any confusion, we make a clear distinction between two types of fixed costs, namely, f_A (abatement investment) and f_I (fixed cost of importing). Although both f_A and f_I are fixed costs, they are determined at their intensive and extensive margins, respectively. Based on the level of productivity and the decision to import (and export, as we will describe later), each firm chooses a level of f_A that leads to the most desired level of abatement discount factor $h(f_A)$. The abatement investment cost is fixed in the sense that it is independent of the scale of production, although different firms choose different levels of abatement investment endogenously. In contrast, the fixed cost of importing f_I is the same for all firms that make import decisions.[23]

Let $\mu^D(\phi)$ and $\mu^M(\phi)$ denote the marginal cost of production for a domestic firm and that for an importing firm, respectively. Cost minimization leads to the following expressions for the marginal cost of production:

$$\mu^D(\phi) = \frac{1}{\phi h(f_A^D(\phi))^{\alpha v_l}} \cdot \tilde{\mu}, \tag{7}$$

$$\mu^M(\phi) = \frac{1}{\phi h(f_A^M(\phi))^{\alpha v_l}} \cdot \tilde{\mu}^*, \tag{8}$$

where we introduce $\tilde{\mu}$ and $\tilde{\mu}^*$ to consolidate the constant terms in the expressions.[24] Note that for given levels of productivity ϕ and abatement investment f_A, the only difference in marginal cost associated with importing arises from the difference between $\tilde{\mu}^*$ and $\tilde{\mu}$. The following lemma shows that a firm can always lower its marginal cost by becoming an importer of foreign intermediate inputs.

Lemma 1. *For given levels of productivity ϕ and abatement investment f_A, the marginal cost of an importing firm is lower than what it would be if the firm were not importing. Namely,*

$$\tilde{\mu}^* < \tilde{\mu}. \tag{9}$$

[22] The fixed cost of abatement investment and that of importing (and exporting in Section 2.4) are expressed in terms of labor units entailed for each corresponding service. Since the wage w is pegged to the price of the numéraire good, which is normalized to 1 in our model, we omit multiplying fixed costs by w.

[23] It is possible to extend our model to include an intensive margin choice of f_I à la Arkolakis (2010), but we do not consider this option in our paper since this feature is not central to our analysis.

[24] In specific,

$$\tilde{\mu} \equiv \left(\frac{w}{(1-\alpha)v_l}\right)^{(1-\alpha)v_l} \left(\frac{\tau}{\alpha v_l}\right)^{\alpha v_l} \left(\frac{c_d}{v_m}\right)^{v_m} v_d^{-\frac{\eta v_m}{\eta-1}},$$

and

$$\tilde{\mu}^* \equiv \left(\frac{w}{(1-\alpha)v_l}\right)^{(1-\alpha)v_l} \left(\frac{\tau}{\alpha v_l}\right)^{\alpha v_l} \left\{\frac{1}{v_m}\left[v_d^{\eta} c_d^{1-\eta} + v_f^{\eta} c_f^{1-\eta}\right]^{\frac{1}{1-\eta}}\right\}^{v_m}.$$

The import decision is made endogenously in our model, and thus some readers may be confused with the concept of "effects of importing" in our discussion. We identify the effects of importing by comparing an importing firm's level of a variable of interest with that under the counterfactual case in which this firm chooses not to import.

Using this definition, Lemma 1 shows that importing lower firms' marginal cost (an effect of importing) and thus if the fixed cost of importing is not a concern, all firms would prefer to import foreign intermediate inputs. The intuition is easily obtained from (5), where the production function features "gains from varieties" of domestic and foreign intermediate inputs.

Firms' profit maximization leads to expressions for the equilibrium prices of their final goods. Recall that firms face an iso-elastic demand with the elasticity given by σ. Profit-maximizing prices for a domestic and an importing firm are, respectively, given by

$$\begin{aligned} p^D(\phi) &= \frac{\sigma}{\sigma-1}\mu^D(\phi), \\ p^M(\phi) &= \frac{\sigma}{\sigma-1}\mu^M(\phi), \end{aligned} \tag{10}$$

which lead to the following expressions for the maximized profits as a function of f_A:

$$\begin{aligned} \pi^D(\phi) &= \frac{1}{\sigma}\left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} EP^{\sigma-1}\tilde{\mu}^{1-\sigma}\left[\phi h(f_A^D(\phi))^{\alpha v_l}\right]^{\sigma-1} - f_A^D(\phi), \\ \pi^M(\phi) &= \frac{1}{\sigma}\left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} EP^{\sigma-1}\tilde{\mu}^{*1-\sigma}\left[\phi h(f_A^M(\phi))^{\alpha v_l}\right]^{\sigma-1} - f_A^M(\phi) - f_I. \end{aligned} \tag{11}$$

Next, we characterize the optimal levels of abatement investment f_A that maximize the profits in expression (11). To this end, we introduce the following parametric specification of $h(f_A)$:

$$h(f_A) \equiv f_A^{\frac{\rho}{\alpha v_l}}, \tag{12}$$

where $\rho > 0$ governs the elasticity of abatement investment. We normalize the exponent by $1/\alpha v_l$ to simplify the notation in Eq. (11). With Eq. (12), the first-order condition with respect to f_A in (11) leads to the following closed-form solutions for f_A[25]:

$$\begin{aligned} \left[f_A^D(\phi)\right]^{1-\rho(\sigma-1)} &= \left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\phi^{\sigma-1}\rho\tilde{\mu}^{1-\sigma}, \\ \left[f_A^M(\phi)\right]^{1-\rho(\sigma-1)} &= \left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\phi^{\sigma-1}\rho\tilde{\mu}^{*1-\sigma}. \end{aligned} \tag{13}$$

For the second-order condition regarding f_A to be valid, we assume that $1-\rho(\sigma-1) > 0$ throughout the paper. A comparison of the two expressions in Eq. (13) immediately leads to the following proposition.

Proposition 1. *For a given level of productivity ϕ, the level of abatement investment for an importing firm is higher than what it would be if the firm were not importing.*

The proof of Proposition 1 follows immediately from (13). The first line of (13) is a domestic firm's abatement investment and the second line is that of an importing firm. Note that the right-hand sides of Eq. (13) show that for a given level of innate firm productivity ϕ, the importing firm invests more in abatement ($f_A^M(\phi) > f_A^D(\phi)$), because we have shown that $\tilde{\mu}^* < \tilde{\mu}$.

Proposition 1 shows that importing increases the firm's abatement investment and its intuition is as follows. As mentioned earlier, abatement investment features an IRS property, such that a larger scale of production makes the abatement investment more worthwhile. At a given level of innate productivity ϕ, an importing firm lowers its marginal cost of production by importing foreign intermediate inputs and produces more output than a domestic counterpart. As such, a firm's import decision leads to an expansion of its scale and an increment in the level of abatement investment.

Our next proposition shows that the choice of f_A amplifies the cost advantage associated with importing.

Proposition 2. *For a given level of productivity ϕ, a higher level of abatement investment associated with importing amplifies the inherent marginal cost advantage associated with importing. That is,*

$$\frac{\mu^M(\phi)}{\mu^D(\phi)} < \frac{\tilde{\mu}^*}{\tilde{\mu}}.$$

Proof. Using Eqs. (7) and (13),

$$\mu^M(\phi) = \left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}^*}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}},$$

[25] Alternatively, it can be shown that

$$f_A^D(\phi) = \frac{\sigma-1}{\sigma}\rho \cdot pq.$$

Thus the operating profit of a non-importing firm is

$$\pi = \frac{1-\rho(\sigma-1)}{\sigma}pq,$$

which is positive as we assume $1-\rho(\sigma-1) > 0$.

$$\mu^D(\phi) = \left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}} \left(\frac{\tilde{\mu}}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}}.$$

Thus,

$$\frac{\mu^M(\phi)}{\mu^D(\phi)} = \left(\frac{\tilde{\mu}^*}{\tilde{\mu}}\right)^{\frac{1}{1-\rho(\sigma-1)}} < \frac{\tilde{\mu}^*}{\tilde{\mu}}. \tag{14}$$

The last inequality follows from $0 < 1 - \rho(\sigma - 1) < 1$. □

The intuition behind Proposition 2 is that a firm's import decision increases the level of abatement investment and thereby lowers the emission tax per unit of output. This channel amplifies the initial cost advantage from importing. Another way to gain intuition regarding Proposition 2 is to observe the production functions in Eqs. (5) and (6). Specifically, abatement investment f_A ameliorates the firm's productivity (note that the term $h(f_A)^{\alpha v_l}$ is isomorphic to a Hicks-neutral productivity term) by increasing the investment in f_A. As we have shown in Proposition 1, an importing firm invests more in f_A and thus improves its productivity to a greater extent.

After characterizing the optimal level of abatement investment, we finally consider the firms' extensive margin decision, namely, whether to import foreign intermediate inputs. To this end, we express the "solved-out" operating profits of a domestic and an importing firm, respectively, as

$$\begin{aligned}\pi^D(\phi) &= \frac{1-\rho(\sigma-1)}{\rho(\sigma-1)}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\tilde{\mu}^{1-\sigma}\phi^{\sigma-1}\rho\right]^{\frac{1}{1-\rho(\sigma-1)}},\\ \pi^M(\phi) &= \frac{1-\rho(\sigma-1)}{\rho(\sigma-1)}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\tilde{\mu}^{*1-\sigma}\phi^{\sigma-1}\rho\right]^{\frac{1}{1-\rho(\sigma-1)}} - f_I.\end{aligned} \tag{15}$$

It is profitable for a firm to import if

$$\pi^M(\phi) > \pi^D(\phi).$$

This condition leads to the following characterization of the importer cutoff productivity.

Proposition 3. *More productive firms import intermediate inputs. Specifically, firms become importers if their productivity level ϕ is greater than ϕ^*, where ϕ^* is given by*

$$\phi^{*\frac{1-\sigma}{1-\rho(\sigma-1)}} = \frac{1}{f_I}\frac{1-\rho(\sigma-1)}{\rho(\sigma-1)}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\rho\right]^{\frac{1}{1-\rho(\sigma-1)}}\left(\tilde{\mu}^{*\frac{1-\sigma}{1-\rho(\sigma-1)}} - \tilde{\mu}^{\frac{1-\sigma}{1-\rho(\sigma-1)}}\right). \tag{16}$$

Proposition 3 predicts a pattern of import decisions stratified by firms' innate productivity, whereby more productive firms engage in international trade.

Having fully solved the firms' problem, we characterize the equilibrium emission intensities of domestic and importing firms. Conventionally, we define emission intensity ζ as the ratio of emissions to total real output.[26] The following proposition shows that an importing firm's emission intensity is lower than that under the constrained equilibrium in which the firm is not allowed to import.

Proposition 4. *For a given level of productivity ϕ, the emission intensity of an importing firm is lower than what it would be if the firm were not importing. Among firms with the same trading statuses, more productive firms exhibit lower emission intensities.*

Proof.

For a domestic firm, the emission intensity can be expressed as

$$\begin{aligned}\zeta^D(\phi) &= \frac{z}{q}\\ &= \frac{1}{\tau}\frac{\alpha v_l(\sigma-1)}{\sigma}\frac{qp}{q}\\ &= \frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}}.\end{aligned} \tag{17}$$

Similarly, for an importing firm:

$$\begin{aligned}\zeta^M(\phi) &= \frac{z}{q}\\ &= \frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}^*}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}}\end{aligned} \tag{18}$$

[26] This definition of emission intensity is consistent with Forslid et al. (2018) and Shapiro and Walker (2018).

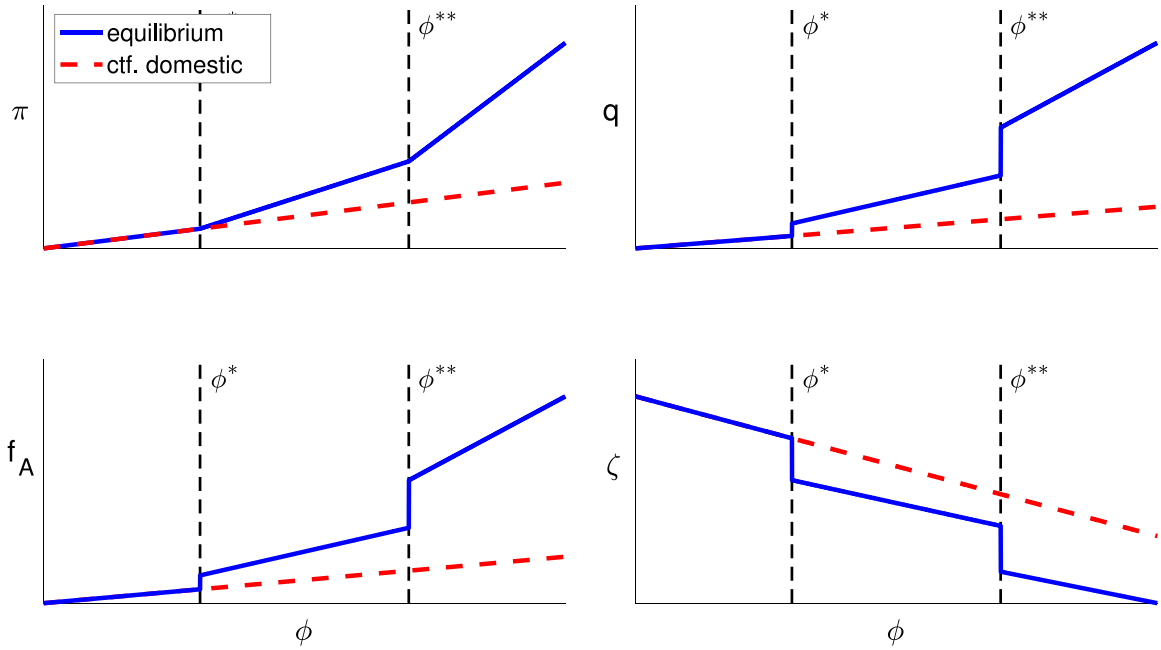

Fig. 2. Equilibrium of key firm-level variables as a function of productivity.
Notes: The blue lines in the figures illustrate the equilibrium levels of key firm-level variables, namely, profit, output, abatement investment and emission intensity (denoted by π, q, f_A and ζ, respectively). The dashed lines in red indicate their counterfactual levels if firms do not engage in international trade. ϕ^* is the importer cutoff productivity, above which firms import intermediate inputs. ϕ^{**} is the exporter cutoff productivity (that we introduce in Appendix A.1), above which firms additionally export final goods. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Comparing Eqs. (17) and (18), it is then clear that

$$\zeta^D(\phi) > \zeta^M(\phi),$$

as we have shown that $\tilde{\mu} > \tilde{\mu}^*$ and $1 - \rho(\sigma - 1) > 0$

The second part of the proposition follows from the observation that both ζ^D and ζ^M are decreasing in ϕ. □

Proposition 4 is key to our analysis and suggests that firms' import decisions reduce their emission intensities, even when differences in productivity levels are accounted for. That is, firms that endogenously choose to import intermediate inputs feature a higher level of productivity and more investment in abatement compared with domestic firms. However, importing firms would invest more in abatement compared with the counterfactual equilibrium in which they were not allowed to import intermediate inputs. In this sense, our model demonstrates that importing reduces firm-level emission intensities.

It is noteworthy that the import channel emphasized in our theoretical framework complements the export channel that has been studied in Forslid et al. (2018). We posit that understanding both import- and export-side impacts of trade is crucial to understand how globalization impacts the environment.

We summarize our key findings in this section in Fig. 2. The blue lines in the figures show that there is a continuous increase in profit around the importer productivity cutoff ϕ^*. However, there are sudden jumps in the levels of output and abatement investment, and a sudden drop of emission intensity around ϕ^*. The discontinuity is driven by the extensive margin decision of importing, which lowers the marginal cost of production from $\mu^D(\phi^*)$ to $\mu^M(\phi^*)$. Profit, in contrast, exhibits a continuous increase around the importer cutoff ϕ^*, since import decision involves a payment of fixed cost f_I. The red dashed lines in the figure illustrate the hypothetical levels of the same variables if the firm in question does not engage in international trade. Fig. 2 clearly shows that importing lowers firms' emission intensities.

2.4. Model extension and discussion

2.4.1. Export decisions

In Appendix A.1, we extend our baseline model to allow firms to make export decisions in addition to import and abatement decisions. It is quite straightforward to show that all of our Propositions 1 through 4 remain valid in the extended model and the additional analysis in Appendix A.1 derives two additional propositions regarding firms' exporting behavior: (i) there is a productivity cutoff, above which firms export their final products à la Forslid et al. (2018) (Proposition A.1) and (ii) exporting decision induces further abatement investment and lowers emission intensities at the firm level (Proposition A.2).

Under an innocuous assumption that the export productivity cutoff is higher than that of importing, we can easily demonstrate the relationship between key outcome variables, import and export decisions and firm productivity in Fig. 2. Note that the lower-left panel shows that there is a jump in firms' abatement investments (blue solid line) once the productivity is slightly above ϕ^{**}, the productivity cutoff for exporting, much greater than not exporting (red dashed line). Accordingly, in the lower-right panel, we observe a significant drop in firms' emission intensity once the productivity surpasses ϕ^{**}. Overall, Fig. 2 depicts how firms' importing and exporting decisions interact with firms' abatement decisions to attain lower emission intensities at the firm level.[27]

2.4.2. Implications on the economy-wide total emissions and further discussions

Implications on the economy-wide total emissions in our model can also be characterized by introducing a balanced trade condition to close the model. However, since the focus of our paper is to study firm-level emission intensities, we relegate discussions on total emissions to Appendix A. We show that the total emissions in this economy are closely tied to its total output, and thus, it is essential to characterize the change in total output in response to trade liberalization to gauge its impact on the environment.

3. Data and empirical strategy

In this section, we describe our main datasets and the key empirical strategy. Our theoretical framework leads to a theory-based regression specification that can be estimated with firm-level data. We also describe our instrument variable strategy to address the endogeneity problem associated with importer and exporter statuses.

3.1. Data description

Firm-level environmental measures are from China's Environmental Statistical Database (CESD) assembled by the Ministry of Ecology and Environment.[28] For each year and a county, top firms that cumulatively account for 85 percent of total pollutants are included in the CESD data. The major pollutants include chemical oxygen demand (COD), oxides of nitrogen (NOX), sulfur dioxide (SO_2), etc. The CESD also collects firms' basic information including name, organization code, administrative division code based on firm location, etc. The CESD dataset is the most extensive nationwide environmental survey in China and has been widely used in studies related to China's environmental issues (e.g., Liu et al. (2017), Wang et al. (2018) and He et al. (2020)).[29]

Firm-level production data are obtained from the Annual Survey of Industrial Enterprises (ASIE) collected and maintained by China's National Bureau of Statistics. This dataset covers all state-owned enterprises (SOEs) and non-SOEs with annual sales over 5 million RMB. The ASIE is the most comprehensive micro-level database for China's manufacturing firms. It reports firm-level information from three major accounting statements (balance sheet, profit statement, and cash flow statement) as well as firms' basic information including the name, organization code, telephone number and zip code, etc.

Transaction-level trade data is from China's General Administration of Customs. China's customs database (CUST) records all import and export transactions as well as the name, telephone number, and zip code of every firm. The CUST dataset provides important information on firms' imports and exports, which helps us to identify the firms' trading statuses in the international market. The CUST also allows us to calculate firms' trade shocks depending on their import and export composition.

Additionally, we use the Trade Analysis Information System (TRAINS) dataset provided by World Integrated Trade Solution (WITS) to obtain product-level trade and tariff data, which are needed to create product-level exogenous shocks.

3.1.1. Merged dataset

For our analysis, we need to merge CESD, ASIE and CUST datasets to obtain a comprehensive set of information for each Chinese firm. In this section, we briefly describe the key steps of the matching procedure and defer the details to Appendix B, where we also discuss how we construct a consistent firm identifier across years.

To match the firms across datasets accurately, we perform pairwise matching between the datasets. The ASIE serves as the bridge to connect the CESD and CUST datasets since it has the most connecting information with the other two datasets. We match the ASIE with the CESD through exact matching by the firm name and organization code, and apply "fuzzy" matching by firm name within the same location for the remaining firms. We then match the ASIE with the CUST through exact matching by firm name and telephone number, and apply "fuzzy" matching by firm name within the same zip code for the remaining firms.[30] We summarize our matching success rates in Fig. 3. On average, firms in our merged dataset account for approximately 75 percent of total output and emissions.

[27] In Appendix A.2, we adopt a more realistic set of parameter values to evaluate our key propositions. The qualitative results remain unchanged.

[28] The Ministry of Ecology and Environment was formerly known as the Ministry of Environmental Protection (2008–2018), and the State Environmental Protection Administration (prior to 2008).

[29] The CESD data is generally based on firms' self-report, but local authorities perform spot checks to ensure data accuracy. The CESD data is also the basis for the official environment statistics released by the Ministry of Ecology and Environment. Rodrigue et al. (2022b) confirm that the SO_2 data in CESD is highly correlated with the US satellite data.

[30] We use the "matchit" command in Stata to calculate similarity scores of firm name pairs based on their "generalized edit distance". The similarity score ranges from 0 to 1 with 1 referring to identical firm names. We set the threshold for firm name pairs that pass the fuzzy matching at 0.7.

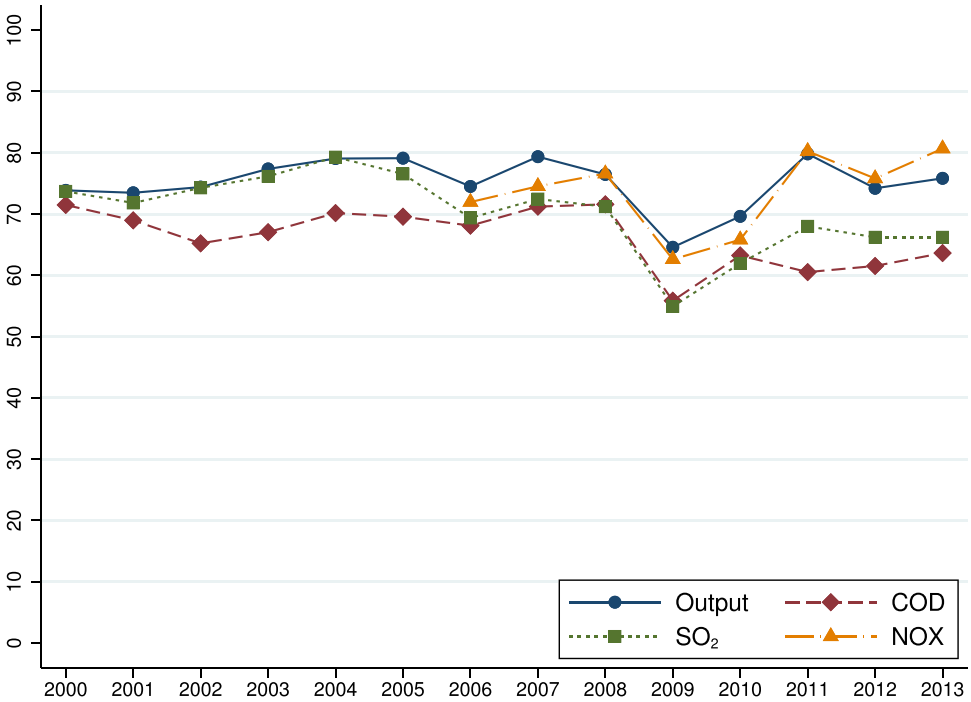

Fig. 3. Percentage of matched output and emissions.
Notes: This figure shows the percentage of output and emissions accounted for by our firm sample for which we have information from both CESD and ASIE datasets. The matching procedure between the two datasets is described in Appendix B.

Table 2
Summary Statistics for the Baseline Regression.

Type	$\log(\zeta^o)$			$\log(\phi)$	τ_X^{IV}	τ_M^{IV}	D_X^{IV}	D_M^{IV}	Obs
	COD	SO_2	NOX						
Domestic	−9.17	−8.01	−9.09	4.15	4.18	3.90	0.04	0.04	297433
Import-only	−10.00	−9.71	−10.26	4.34	2.69	3.01	0.55	0.64	96841
Export-only	−9.95	−9.60	−10.41	4.27	2.48	3.25	0.63	0.54	99170
Both	−10.11	−9.93	−10.58	4.31	2.32	3.00	0.67	0.67	71210

Notes: This table presents the sample averages of key variables in the baseline IV regressions in Table 5. Domestic firms refer to the firms that do not import or export goods in a given year. $\log(\zeta^o)$ denotes the real output-based emission intensity at the firm level, $\log(\phi)$ denotes the labor productivity, and $\{\tau_X^{IV}, \tau_M^{IV}, D_X^{IV}, D_M^{IV}\}$ are the set of instrument variables for the firms' endogenous importer and exporter statuses described in Section 3.4. The number of observations in the last column represent the number of firm-year observations.

3.2. Summary statistics

The key summary statics for the regression sample are reported in Table 2. We report the mean of the variables by the type of trading firms. First, we notice that domestic firms feature the highest emission intensities on average. The firms that engage in both importing and exporting have the lowest emission intensities across the emission types, suggesting that *both* importing and exporting reduce emission intensities.

Second, firms that only engage in importing feature the highest productivity level on average. This suggests that firms' productivity level is not a sufficient statistic for firms' trading status and we rationalize this in Appendix C by relaxing the assumption that all firms face the same fixed cost of trading. For our instrument variables, we note that "export-only" and "both" firms feature the lowest τ_X^{IV} and the highest D_X^{IV}, whereas "import-only" and "both" firms feature the lowest τ_M^{IV} and the highest D_M^{IV}, as expected. Finally, we note that trading firms in each type account for approximately 15 to 17 percent of total observations, which provides sufficient variations in firms' trading statuses to identify the causal effect of international trade.

3.3. From theory to the estimating equation

It is not uncommon for firms to export final goods without importing any intermediate inputs. Our original theoretical framework can account for these firms with a minor adjustment in the assumption about fixed costs. Specifically, we show in Appendix C that

if we relax the assumption that the fixed costs of importing and exporting are the same across all firms, we can account for all four types of firms (taxonomized by their importer and exporter statuses) in equilibrium. In Appendix C, we further show that the emission intensities of the four types of firms can be structurally decomposed as

$$\begin{aligned}
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) && \text{for } \mathbb{D},\\
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) + \kappa_M && \text{for } \mathbb{M},\\
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) + \kappa_X && \text{for } \mathbb{X},\\
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) + \kappa_M + \kappa_X && \text{for } \mathbb{Y},
\end{aligned} \tag{19}$$

where $\mathbb{D}, \mathbb{M}, \mathbb{X}$ and $\mathbb{Y}$ denote, respectively, the set of domestic, importing, exporting firms and firms that both import and export. κ is a constant term, $-\frac{1}{1-\rho(\sigma-1)}\log(\phi)$ is the reduction of emission intensities from higher productivity, κ_M is a common term for importers, and κ_X is a common term for exporters. For firms that *both* import and export, κ_M and κ_X terms apply additively.[31]

Eq. (19) naturally motivates the following regressions specification :

$$\log(\zeta_{f,t}) = \kappa_M \cdot D_{fM,t} + \kappa_X \cdot D_{fX,t} + \beta_\phi \log(\phi_{f,t}) + \beta_D \mathbf{D} + \epsilon_{f,t}, \tag{20}$$

where $D_{fM,t}$ and $D_{fX,t}$ are dummy variables that are equal to one if the firm f imports or exports in year t, respectively, $\phi_{f,t}$ denotes the level of firm productivity, and $\epsilon_{f,t}$ is the error term.[32] Since we include firm fixed effects, the estimated parameters $D_{fM,t}$ and $D_{fX,t}$ imply the average change of emission intensities if a firm changes its trade status. $\mathbf{D}$ is a vector of covariates defined as $\mathbf{D} \equiv \{\mathbf{D}_f, \mathbf{D}_{i,t}, \mathbf{D}_{p,t}\}$, where $\mathbf{D}_f$ is a set of firm fixed effects, $\mathbf{D}_{i,t}$ is a set of dummy variables for the interactive industry-year fixed effects, and $\mathbf{D}_{p,t}$ is a set of dummy variables for the interactive province-year fixed effects. The term $\mathbf{D}_f$ absorbs any firm-specific time-invariant factors (such as firms' specific know-how to reduce emissions, ownership status, establishment year, etc.) that are not absorbed by the firm productivity, $\mathbf{D}_{i,t}$ can fully absorb the industrial prices, environmental regulation and other unobserved characteristics that vary at the industry-year level, and $\mathbf{D}_{p,t}$ accounts for the provincial level differences in environmental policy stringency that evolve differently over time.[33]

In our baseline specifications, we experiment with two definitions of emission intensities, ζ^o and ζ^v. The first one, ζ^o, conventionally defines firm-level emission intensities as the mass of emissions normalized by the level of *real output*.[34] However, importing firms, holding constant the level of abatement investment f_A, tend to yield greater output, because they use more intermediate inputs than non-importing firms. Hence, we experiment with an alternative definition, ζ^v, in which emission intensities are defined as the mass of emissions normalized by the firms' *real value-added*. Adopting the second definition of emission intensities helps to rule out the simple emission intensity reduction channel that importing firms scale up their output with extensive use of foreign intermediate inputs. In Appendix D, we show that our key propositions regarding firm-level emission intensities are also applicable to those normalized by real value-added. In general, our regression results are robust to both definitions.[35]

[31] Specifically,

$$\kappa \equiv \log\left(\frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\right) - \frac{\rho}{1-\rho(\sigma-1)}\log\left(EP^{\sigma-1}\right) + \frac{1}{1-\rho(\sigma-1)}\log(\tilde{\mu}).$$

The importer term κ_M is defined as

$$\kappa_M \equiv -\frac{1}{1-\rho(\sigma-1)}\left(\log(\tilde{\mu}) - \log(\tilde{\mu}^*)\right),$$

and the exporter term κ_X is defined as

$$\kappa_X \equiv -\frac{\rho}{1-\rho(\sigma-1)}\left\{\log\left[EP^{\sigma-1} + E^*(P^*/\delta)^{\sigma-1}\right] - \log\left(EP^{\sigma-1}\right)\right\}.$$

Some readers might wonder why the volumes of imports and exports do not enter Eq. (19). According to our theoretical framework, greater amounts of trade affect the firm-level emission intensities through the productivity term ϕ. Thus, the effects of greater trade volume are captured by the productivity term.

[32] In our baseline specification, we use labor productivity (value added per worker) as a proxy for the firm-level productivity.

[33] Using Canadian data, Najjar and Cherniwchan (2021) show that environmental regulation is a key factor for firms' emission intensities. The literature on Chinese environmental regulations suggests that province-year and industry-year level fixed effects can be effective to account for policy changes (Wu et al., 2017; Chen et al., 2018; Shi and Xu, 2018). For instance, Kwon et al. (2021b) show that Chinese provinces faced varying emission reduction targets during the five-year plan that began in 2006.

[34] Previous studies that define firm-level emission intensities as emissions normalized by corresponding levels of total outputs include, but not limited to, Cui et al. (2016), Forslid et al. (2018), Shapiro and Walker (2018) and Rodrigue et al. (2022a). One exception to this is Hering and Poncet (2014), but they focus on emission intensities at the sectoral level, and define the sectoral level emission intensities as the total emissions divided by the aggregate sectoral value-added.

[35] To account for approximately 10% of firms that report zero emissions, we calculate the emission intensities as $\zeta^o = \frac{1+\text{emissions}}{1+\text{output}}$. We add 1 to the numerator so that the dependent variable is well defined after taking a natural logarithm, and we also add 1 to the denominator to reduce the impact of adding 1 to the numerator to the ratio. The simple correlation between $\ln(\zeta^o)$ and $\ln(\frac{\text{emissions}}{\text{output}})$ is over 0.95. Simple regression results yield slope coefficients of around 1.1 and R^2 of 0.95.

3.4. Endogeneity concerns and the IV for importer and exporter statuses

A challenge to estimating Eq. (20) with an ordinary least squares (OLS) is that the firms' importer and exporters statuses are endogenous. Time-varying shocks that are not absorbed by the firm fixed effects potentially bias the estimates of κ_M and κ_X.[36] Moreover, measurement errors associated with firms' importer and exporter statuses may also cause attenuation bias in the OLS estimation.[37] Thus, we propose the following instrument variable approach to overcome the endogeneity issues.

The first set of our IV is akin to Bombardini et al. (2019). Specifically, let $\tau_{X,h}$ and $\tau_{M,h}$ denote the average tariffs faced by Chinese exporters and importers of an HS 6-digit product h, respectively.[38] Both $\tau_{X,h}$ and $\tau_{M,h}$ are weighted by the trade volumes between China and the respective trading partners. Similar to Bombardini et al. (2019), the first set of IVs is defined as

$$\tau_{fi,t}^{IV} = \sum_h \frac{v_{fih,t-1}}{v_{fi,t-1}} \tau_{ih,t} \quad \text{for } i \in \{X, M\}, \tag{21}$$

where $v_{fih,t-1}$ denotes the firm f's trade volume of product h in direction $i \in \{X, M\}$ (X and M denote exports and imports, respectively) in year $t-1$, $v_{fi,t-1}$ is the firm's total volume of trade in direction i, and $\tau_{ih,t}$ is the tariff level of product h in direction i.[39] Intuitively, $\tau_{fi,t}^{IV}$ captures the tariffs faced by the Chinese firm f in trade direction i weighted by $\frac{v_{fih,t-1}}{v_{fi,t-1}}$, which is its trade share of product h in the preceding year.[40]

The second set of our IV applies the shift-share idea popularized in the international trade literature by Autor et al. (2013). Specifically, let $\mathcal{X}_{h,t}$ denote the total volume of trade in product h between countries in the rest of the world.[41] The second set of IVs for importer and exporter statuses is given by

$$D_{fi,t}^{IV} = \sum_h \frac{v_{fih,t-1}}{v_{fi,t-1}} \frac{\mathcal{X}_{h,t}}{\mathcal{X}_{h,t-1}} \quad \text{for } i \in \{X, M\}. \tag{22}$$

That is, $D_{fi,t}^{IV}$ captures the firm's exposure to the trade shocks from the rest of the world, $\frac{\mathcal{X}_{h,t}}{\mathcal{X}_{h,t-1}}$, weighted by its trade share of product h in the preceding year, $\frac{v_{fih,t-1}}{v_{fi,t-1}}$. Intuitively, if there is no aggregate shock in trade in year t ($\mathcal{X}_{h,t} = \mathcal{X}_{h,t-1}$), the value of $D_{fi,t}^{IV}$ is 1. Namely, the firm that used to import (or export) in year $t-1$ is expected to continue to do so if there are no aggregate shocks. However, if there are negative trade shocks for some products, it will lead to a value of $D_{fi,t}^{IV}$ that is less than 1 depending on the firm's trade shares.[42] In the first stage regressions, we expect that $\tau_{fi,t}^{IV}$ to be *negatively* (and $D_{fi,t}^{IV}$ to be *positively*) correlated with $D_{fi,t}$ for $i \in \{X, M\}$.

The key working assumption of our IV strategy can find its root from a large body of literature (e.g., Autor et al. (2013), Hummels et al. (2014) Bombardini et al. (2019)). The novelty of our approach is to adopt an over-identifying instrument strategy where four instrument variables ($\tau_{fi,t}^{IV}$ and $D_{fi,t}^{IV}$, $i \in (X, M)$) are jointly used to improve the statistical efficiency. The first component of the IV, a firm's trade shares in the previous year ($v_{fih,t-1}/v_{fi,t-1}$), is most likely to affect a firm's emission intensity through the firm's trading statuses or its productivity level in the current year. This assumption is analogous to that of De Loecker (2013) where the current productivity level of the firm becomes a sufficient statistic for the past exporting experience. The second component of our IV, the aggregate foreign demand shock for product h, can also be regarded as exogenous to individual firms. The last component, the tariff changes $\tau_{ih,t}$, is also exogenous to individual firms, but they may be correlated with unobserved environmental policy (Shapiro, 2021). To alleviate this concern, we include a battery of province-year and industry-year fixed effects, which can absorb environmental policies that vary at either industry or province level.[43]

Recent papers have re-examined the nature of the shift-share instruments and allowed us to examine the validity of such instruments.[44] In Appendix E, we provide a careful treatment of our instrument variables. In specific, we convincingly show that our shocks are not correlated with the error term in the second stage and that our IV regression results are not driven by abnormal shocks in particular subset of HS products.

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36 For example, more stringent environment regulations make it more difficult for Chinese firms to compete with foreign firms in the export market (Shi and Xu, 2018), or local environment protection policies may be correlated with export-promoting (or import-restricting) policies (Shapiro, 2021). Our province-year and industry-year fixed effects may still be insufficient to capture all unobserved characteristics.

37 Some firms have insignificant amounts of export or import transactions relative to their sizes. We posit that the typical effects of international trade would not be reflected in these firms.

38 $\tau_{X,h}$ is set by China's trading partners and $\tau_{M,h}$ is set by the Chinese government.

39 For non-importing or non-exporting firms in year $t-1$, $v_{fi,t-1}$ equals to 0, and thus Eq. (21) is not defined. Thus, for non-importing or non-exporting firms, we impose the highest level of tariffs faced by the importing or exporting firms from the same industry.

40 Shapiro (2021) shows that governments may use fiscal policies (e.g., tariffs) to induce better environmental outcomes. We note that the reason behind tariff changes is not a concern for our identification strategy as long as they are exogenous to individual firms. As we discussed earlier, environmental policies, which may be correlated with tariff changes, are controlled with province-year and industry-year fixed effects.

41 Note that total exports from the rest of the world to the rest of the world are the same as total imports, by definition.

42 For domestic firms in year $t-1$, we set $D_{fi,t}^{IV}$ to be 0. The idea is that firms that did not use to engage in international trade in year $t-1$ are naturally less exposed to aggregate trade shocks.

43 Studies have shown that COD and SO_2 emission regulations varied mainly at the province level (see Shi and Xu (2018) and Kwon et al. (2021b)).

44 Leading papers in this thread of literature include (but are not limited to) Adao et al. (2019), Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022). Campante et al. (2023) provide a case of comprehensive application of these methods in practice.

Table 3
Effects of International Trade on Emission Intensities (OLS).

Dep. Var.	(1)	(2)	(3)	(4)	(5)	(6)
Emis. Type	$\log(\zeta^o)$			$\log(\zeta^v)$		
	COD	SO_2	NOX	COD	SO_2	NOX
Panel A. without controlling for productivity						
κ_M (importer)	−0.0822***	−0.0715***	−0.0463***	−0.0766***	−0.0674***	−0.0420***
	(0.00781)	(0.00891)	(0.00984)	(0.00835)	(0.00945)	(0.0107)
κ_X (exporter)	−0.0561***	−0.0695***	−0.0431***	−0.0352***	−0.0458***	−0.0228**
	(0.00756)	(0.00868)	(0.00966)	(0.00815)	(0.00928)	(0.0105)
Adjusted R^2	0.770	0.808	0.816	0.738	0.774	0.777
p-value ($\kappa_M = \kappa_X$)	0.02	0.88	0.82	0.00	0.12	0.21
Panel B. controlling for productivity						
κ_M (importer)	−0.0713***	−0.0595***	−0.0436***	−0.0547***	−0.0428***	−0.0357***
	(0.00749)	(0.00865)	(0.00960)	(0.00739)	(0.00858)	(0.00965)
κ_X (exporter)	−0.0552***	−0.0674***	−0.0448***	−0.0332***	−0.0411***	−0.0265***
	(0.00726)	(0.00839)	(0.00939)	(0.00721)	(0.00834)	(0.00938)
β_ϕ (productivity)	−0.347***	−0.344***	−0.256***	−0.708***	−0.708***	−0.624***
	(0.00386)	(0.00403)	(0.00483)	(0.00409)	(0.00428)	(0.00558)
Adjusted R^2	0.785	0.819	0.823	0.797	0.822	0.823
p-value ($\kappa_M = \kappa_X$)	0.15	0.53	0.93	0.05	0.90	0.50
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	422234	397146	215483	422326	397238	215579
Firms	69551	65488	47139	69556	65489	47143

Notes:
(i) This table shows the OLS regression results of firm-level emission intensities on importer and exporter statuses. The regression specification is presented in Eq. (20). ζ^o and ζ^v denote the firm-level emission intensities normalized by the value of real total output and value-added, respectively. κ_M and κ_X are the coefficients of the dummy variables for the importer and exporter statuses respectively. β_ϕ is the coefficient of the natural logarithm of labor productivity. p-value reports the result of the t-test where the null hypothesis assumes $\kappa_M = \kappa_X$.
(ii) Panels A and B show the regression results without and with firm-level productivity in the regressors, respectively.
(iii) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

4. Empirical results

This section presents our main regression results. We first use the OLS regression results to document key data patterns. This is followed by the first stage regressions and our IV regression results. We also use data to directly test the abatement channel that we propose in our theory section. Finally, we adopt a novel approach to eliminate the competing effects from the POH.

4.1. OLS regression results

As a first step, we present the OLS regression results in Table 3. The dependent variables in columns (1) through (3) are emission intensities normalized by total output, and those in columns (4) through (6) are emission intensities normalized by value-added. Regression results in columns (1) through (3) (and those in columns (4) through (6)) respectively study different types of emissions. Due to data availability, our regression samples differ depending on the dependent variable. Specifically, the NOX emissions were not reported in the CESD data until 2006. Thus, the regression results in columns (3) and (6) are based on a shorter panel sample.

Panels A and B are separate regressions using the same firm sample, where regression results in Panel A do not control for the firms' productivity in year t, and those in Panel B do so. All regression specifications include firm, province-year and industry-year fixed effects.

Consistent with our theory, both regression results reported in panels A and B indicate that the estimates of κs are negative and highly significant. Panel A of Table 3 shows that importing (exporting) firms exhibit 2 percent to 8 percent lower emission intensities compared with otherwise comparable non-importing (non-exporting) firms. In both panels A and B, we report the p-values of the two-sided t-tests with the null hypothesis being $\kappa_M = \kappa_X$. We observe that the estimates of κ_M are significantly greater in magnitude than κ_X for COD, but they are not statistically discernible for other emission types in general.

Comparing the results in panels A and B, the estimates of κ_M and κ_X are very similar regardless of whether the productivity term is explicitly controlled for. This can be understood with the supplementary regression results shown in Table G.1 in Appendix G that 73 percent variation in firm-level labor productivity is explained by the fixed effects. Thus, it is plausible that further controlling for the firm-level productivity in addition to the fixed effects do not affect estimates of κs significantly.

Table 4
First stage regression results.

Regression sample	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	COD		SO_2		NOX	
	D_M	D_X	D_M	D_X	D_M	D_X
τ_X^{IV}	0.000116***	−0.000144***	9.55e−05***	−0.000143***	7.40e−05***	−0.000181***
	(1.93e−05)	(2.10e−05)	(2.17e−05)	(2.32e−05)	(2.87e−05)	(3.02e−05)
τ_M^{IV}	−0.00299***	7.30e−05	−0.00283***	0.000273***	−0.00351***	0.000288***
	(6.37e−05)	(6.29e−05)	(7.31e−05)	(7.11e−05)	(8.27e−05)	(8.42e−05)
D_X^{IV}	−0.0263***	0.118***	−0.0240***	0.124***	−0.0487***	0.0393***
	(0.00311)	(0.00329)	(0.00348)	(0.00367)	(0.00443)	(0.00465)
D_M^{IV}	0.0144***	−0.0482***	0.0201***	−0.0404***	−0.0616***	−0.0474***
	(0.00351)	(0.00344)	(0.00405)	(0.00393)	(0.00431)	(0.00438)
β_ϕ (productivity)	0.00410***	0.00129*	0.00397***	0.00162**	0.00126	−0.000672
	(0.000726)	(0.000738)	(0.000687)	(0.000699)	(0.000978)	(0.000982)
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	422234	422234	397146	397146	215483	215483
Firms	69551	69551	65488	65488	47139	47139

Notes:
(i) This table presents the first-stage regression results for IV regressions. Columns (1) and (2) report the first stage regression results for the IV regressions in column (1) of Panel B in Table 5. Columns (3) and (4) are for the IV regression results in column (2) of the same panel. Columns (5) and (6) are for the IV regression results in column (3) of the same panel.
(ii) All regressions include fixed effects at the firm, province-year and industry-year levels.
(iii) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

We take results in Table 3 to be preliminary evidence that firms' international trade statuses are a key determinant of emission intensities. These results account for any time-invariant firm characteristics and annual productivity shocks. Yet, concerns with reverse causality and measurement errors associated with the importer and exporter statuses remain. These concerns are addressed in the IV regression results in the following subsections.

4.2. First stage regressions

Before discussing the main IV regression results, we present our first-stage regression results in Table 4 to show the relevance of our instruments with the endogenous variables. As mentioned earlier, our regression samples differ by the type of emissions due to data availability. Thus, the first-stage regression results for COD, SO_2 and NOX are separately reported in Table 4.[45]

In Table 4, we confirm our expected signs of IVs in the first stage regressions. In general, we observe that D_is are predicted negatively by τ_i^{IV}s and positively by D_i^{IV}s for $i \in \{M, X\}$.[46] One interesting observation, albeit not central to our identification strategy, from Table 4 is that τ_j^{IV}s and D_j^{IV}s ($j \neq i$) tend to predict D_i with the opposite signs. This may arise if aggregate shocks (or tariffs) on products that Chinese firms typically import and export are negatively correlated. (e.g., the global trade volume of rubber increases while that of tires decreases.)

4.3. IV regression results

Our baseline IV regression results are presented in Table 5. Similar to the OLS results in Table 3, regression results are presented in different columns and panels depending on their dependent variables and whether the firm-level productivity is included in the regression.

IV regression results convincingly support our hypothesis that both importer and exporter statuses cause a significant reduction in firm-level emission intensities. Specifically, columns (1) through (3) in panel A of Table 5 show that both importing and exporting led to significant reduction in COD, SO_2 and NOX emission intensities with the effects ranging from 15 to 38 percent. The p-values presented at the bottom of both panels suggest that importing has comparable negative effects on emission intensities as exporting for the majority of cases. The F-statistics at the bottom of both panels suggest that our IVs do not suffer from a weak-IV problem. The results in panel B are similar, suggesting that conditional on fixed effects, inclusion of the firm-level productivity term does

[45] In Table 4, we report the first stage results where the productivity term is an included exogenous variable. The first stage regression results without the productivity term (available on request) are very similar.

[46] The only one exceptions are the coefficients of D_M^{IV} in columns (3) and (5). In the CESD dataset, NOX emissions are not reported until 2006, and thus the data panel to study the NOX emissions is shorter. As shown from the F-statistics reported in Table 5, our IVs do not perform equally well for the NOX regressions. The structure of a shorter panel means that a greater proportion of the variation in the dependent variables is absorbed by the firm fixed effects, making it difficult to identify the coefficients of time-varying covariates.

Table 5
Effects of international trade on emission intensities (IV).

Dep. Var.	(1)	(2)	(3)	(4)	(5)	(6)
Emis. Type	$\log(\zeta^o)$			$\log(\zeta^v)$		
	COD	SO_2	NOX	COD	SO_2	NOX
Panel A. without controlling for productivity						
κ_M (importer)	−0.166***	−0.201***	−0.384***	−0.140***	−0.190***	−0.379***
	(0.0393)	(0.0449)	(0.0585)	(0.0423)	(0.0479)	(0.0660)
κ_X (exporter)	−0.207***	−0.228***	−0.258**	−0.153***	−0.165***	−0.254*
	(0.0501)	(0.0546)	(0.119)	(0.0543)	(0.0588)	(0.133)
Kleibergen–Paap F	502.89	423.11	79.03	502.70	423.01	78.91
p-value ($\kappa_M = \kappa_X$)	0.48	0.67	0.22	0.84	0.71	0.27
Panel B. controlling for productivity						
κ_M (importer)	−0.153***	−0.173***	−0.391***	−0.113***	−0.133***	−0.398***
	(0.0381)	(0.0437)	(0.0569)	(0.0376)	(0.0430)	(0.0578)
κ_X (exporter)	−0.199***	−0.214***	−0.259**	−0.138***	−0.136***	−0.262**
	(0.0485)	(0.0531)	(0.117)	(0.0481)	(0.0527)	(0.119)
β_ϕ (productivity)	−0.347***	−0.343***	−0.256***	−0.708***	−0.708***	−0.623***
	(0.00386)	(0.00403)	(0.00483)	(0.00409)	(0.00429)	(0.00558)
Kleibergen–Paap F	502.79	422.96	79.03	502.60	422.86	78.90
p-value ($\kappa_M = \kappa_X$)	0.41	0.51	0.19	0.66	0.95	0.18
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	422234	397146	215483	422326	397238	215579
Firms	69551	65488	47139	69556	65489	47143

Notes:
(i) This table shows the IV regression results of firm-level emission intensities on importer and exporter statuses. The regression specification is presented in Eq. (20). ζ^o and ζ^v denote the firm-level emission intensities normalized by the value of real total output and value-added, respectively. κ_M and κ_X are the coefficients of the dummy variables for the importer and exporter statuses respectively. β_ϕ is the coefficient of the natural logarithm of labor productivity. p-value reports the result of the t-test where the null hypothesis assumes $\kappa_M = \kappa_X$. The instruments for the endogenous variables are discussed in Section 3.4. First stage regression results are presented in Table 4.
(ii) Panels A and B show the regression results without and with firm-level productivity in the regressors, respectively.
(iii) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

not alter the estimates of κs significantly. The results (in columns (4) through (6)) remain robust if we use emission intensities normalized with value-added as dependent variables.

Compared with the OLS results, the coefficient estimates in the IV regression results are greater in magnitude. We believe that this can be explained by the environmental bias in trade policy that causes positive correlation between firms' emission intensities and trading statuses (Shapiro, 2021). This may also be driven by firms marginally involved with international trade. A sizable share of trading firms' import and export volumes are extremely small compared with their total outputs. We suspect that these firms do not engage in international trade as an integral part of their operation. Classifying them as importing or exporting firms naturally create a downward bias (in magnitude) for the estimates of the impact of international trade. We also note that the IV estimates of the exporter effects are in line with the previous findings in the literature (e.g., Holladay (2016), Forslid et al. (2018), and Rodrigue et al. (2022a)).[47]

4.4. Mechanism and competing hypotheses

Our baseline model argues that both engaging in importing and exporting induce more investments in abatement and hence a reduction in firm-level emission intensities. In this subsection, we further exploit the variables in the CESD data to confirm this mechanism. Furthermore, we perform a robustness check to show that our key findings remain unchanged even if we account for the POH — a competing hypothesis that also creates a negative link between a firm's importer status and emission intensities.

[47] We caution the readers from arriving at the conclusion that NOX reduction is more responsive to international trade than other pollutants. First, NOX data becomes available since 2006, and thus the difference in coefficients may simply reflect that firm samples are different. Second, both COD and SO_2 experienced nation-wide abatement campaign during the five-year plan (2006–2010). If the abatement cost is convex, it is possible that the additional impact from international trade may be diminished after a nation-wise increase in the level of abatement.

Table 6
Effects of international trade on abatement (IV).

Dep. Var.	(1)	(2)	(3)	(4)
		Removal of		
	Water treat. cap.	COD	SO_2	NOX
Panel A. without controlling for productivity				
κ_M (importer)	0.701***	1.258***	0.333***	0.159***
	(0.0529)	(0.0812)	(0.0585)	(0.0404)
κ_X (exporter)	0.791***	0.741***	0.398***	0.0870
	(0.0677)	(0.100)	(0.0715)	(0.0799)
Kleibergen–Paap F	478.34	450.08	396.05	78.63
Panel B. controlling for productivity				
κ_M (importer)	0.699***	1.258***	0.332***	0.159***
	(0.0529)	(0.0812)	(0.0585)	(0.0404)
κ_X (exporter)	0.791***	0.741***	0.398***	0.0869
	(0.0677)	(0.100)	(0.0715)	(0.0799)
β_ϕ (productivity)	0.0369***	0.0140***	0.0146***	−0.00429*
	(0.00392)	(0.00526)	(0.00400)	(0.00249)
Kleibergen–Paap F	478.25	450.02	395.91	78.62
Firm FE	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓
Obs.	404153	390987	375730	213632
Firms	68411	67665	64265	46888

Notes:
(i) This table shows the IV regression results of firm-level abatement mechanisms on importer and exporter statuses. The regression specification is similar to Eq. (20). The dependent variable in column (1) is the natural logarithm of the firm-level water treatment capacity (measured in tonnes). The dependent variables in columns (2), (3) and (4) are respectively the natural logarithm of the firm-level emission removals (measured in tonnes). κ_M and κ_X are the coefficients of the dummy variables for the importer and exporter statuses respectively. β_ϕ is the coefficient of the natural logarithm of labor productivity. p-value reports the result of the t-test where the null hypothesis assumes $\kappa_M = \kappa_X$. The instruments for the endogenous variables are discussed in Section 3.4.
(ii) Panels A and B show the regression results without and with firm-level productivity in the regressors, respectively.
(iii) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

4.4.1. Test of abatement mechanisms

The CESD data contain information on the capacity of water treatment at the firm level, which is the variable that corresponds most closely to the variable $h(f_A)$ in our model.[48] The data also contain the masses of COD, SO_2 and NOX end-of-pipe treatment, with which we infer the abatement capacities of these pollutants.[49]

To confirm the mechanism of our model, we follow the main regression specification in Eq. (20), but replace the dependent variables with the (natural logarithm of) mechanism variables. We also instrument both importer and exporter statuses with our IVs to avoid the endogeneity problem. If the estimated κs are positive, we can confirm that international trade leads to the expansion of abatement capacities.

The mechanism regressions are presented in Table 6. We observe that, regardless of whether we include firm-level productivity as an additional control, both importer and exporter statuses significantly raise all the mechanism variables. International trade has especially strong positive effects on water treatment capacities, COD removal and SO_2 removal, whereas the effects are modest for NOX removal.

4.4.2. Accounting for pollution offshoring hypothesis

A competing hypothesis, POH, posits that firms can reduce their own emissions by offshoring parts or the entirety of emission-intensive production tasks abroad.[50] This theory would also predict a negative correlation between firms' importer statuses (but not their exporter statuses) and their emission intensities, calling into question whether our previous empirical results are able to disentangle the POH from our theory.

This subsection attempts to tease out potential complications from the POH. To fix ideas, the classic POH predicts a negative link between importing and emissions if firms outsource emission-intensive tasks (for the production of intermediate inputs) and subsequently import the intermediate inputs as a way to reduce their own emissions (Cherniwchan et al., 2017). In other words,

[48] In our theoretical framework, $h(f_A)$ indicates the discount factor of total emissions associated with an investment level f_A.

[49] Note that emissions abatement can be achieved by end-of-pipe treatment, innovation in production processes or both. The abatement investment in our theoretical framework does not take a particular stance. Constrained by data availability, we test the abatement mechanism with the mass of end-of-pipe treatment.

[50] See Cherniwchan et al. (2017) for the discussion on POH, and see Cherniwchan (2017) for the suggestive evidence of POH.

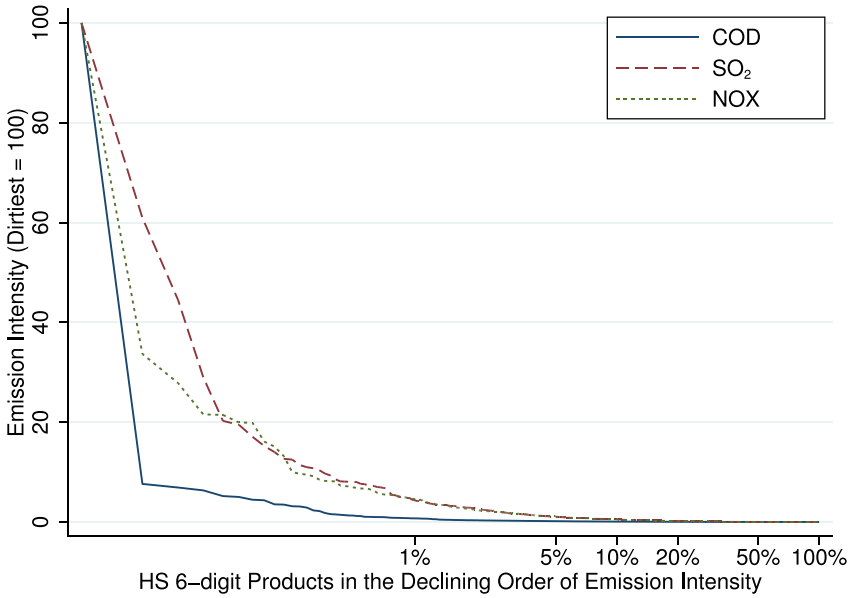

Fig. 4. Discrepancy in emission intensities across HS 6-digit products.
Notes: This figure illustrates the emission intensities of HS 6-digit products. A total of 4,432 HS 6-digit products are aligned in the descending order of their emission intensities along the horizontal axis. For each emission type, we normalize the emission intensity of the dirtiest product as 100.

POH manifests itself mainly through the firms that import "*dirty products*", the production of which generates significant emissions. As such, we drop all the firms that import any "dirty products" in a given year from our regression sample as a plausible way to isolate the effects of the POH.[51]

In Appendix F, we describe the procedure to calculate emission intensities embodied in the production of each HS 6-digit product.[52] To put it succinctly, we use the matched CESD, ASIE and the Chinese customs data to calculate the entailed emissions at the firm-product level by assuming that the emissions associated with each product for a given firm are proportional to the product's output share in the firm's total output.[53]

Using the method described in Appendix F, we are able to rank each HS 6-digit product by their emission intensities. In Table G.6 in Appendix G, we shortlist top 5 most emission-intensive HS 6-digit products in respective emission categories. As expected, we observe that the key emission intensive industries are paper manufacturing, chemical industries and textile industries.

A remarkable pattern emerges from the product-level emission intensity distribution illustrated in Fig. 4. Most strikingly, there is a huge discrepancy in emission intensities. We observe that the emission intensities of the bottom 95 percent of the HS 6-digit products are almost negligible compared with those in top 1 percent.

Motivated by Fig. 4, we define a firm to be a *dirty importer* if the firm imports at least a single product among the top 10 percent emission-intensive HS 6-digit products in a given year.[54] We believe that dropping dirty importers from the regression sample can effectively eliminate the POH channel that creates a negative link between firms' importer status and their emission intensities. In the extreme case, if our baseline regression results are entirely driven by the POH, the estimates of κ_M may not be discernible from 0 in this sub-sample of firms.

Regression results without dirty importers are presented in Table 7. We observe that the estimates of κ_M are still negative and significant. However, compared with the main IV results in Table 5, the estimates of κ_M decline slightly in magnitude except for column (1). This suggests that the effects of POH exist among Chinese firms, but the POH does not entirely account for the negative effects of importing on emission intensities. Table 7 indicates that importing significantly lowered emission intensities for all emission types.

[51] Conceptually, firms can avoid importing "dirty products" while offshoring the pollution-intensive tasks, if they assemble the final products entirely in foreign countries (e.g., Apple entirely outsources the iPhone production, of which the production of semi-conductors and batteries are detrimental to the environment.). Chinese firms rarely employed such production arrangement during the time frame of our study.

[52] We remind readers that our definition of "dirty products" are based on the emissions associated with the production process and not during consumption. We argue that dirty products defined in this way is relevant for us to rule out the competing effects from the POH.

[53] This assumption is equivalent to assuming that the firm's output mix features similar emission intensities. In Kwon et al. (2021a), we develop a more sophisticated method to calculate product level emission intensities by solving for the fixed point of emission intensities that are consistent with how emissions are allocated within firms across products. We note that our regression results are not sensitive to taking either approach.

[54] Our conclusion remains unchanged if we adopt a more conservative definition of dirty importers by switching from importing any of top 10 percent emission-intensive HS 6-digit products to top 20 percent. The alternative regression results are available upon request.

Table 7
Effects of international trade on emission intensities (IV regressions without dirty importers).

Dep. Var.	(1)	(2)	(3)	(4)	(5)	(6)
Emis. Type	$\log(\zeta^o)$			$\log(\zeta^v)$		
	COD	SO_2	NOX	COD	SO_2	NOX
Panel A. without controlling for productivity						
κ_M (importer)	−0.196***	−0.220***	−0.273***	−0.149***	−0.187***	−0.264***
	(0.0432)	(0.0487)	(0.0568)	(0.0465)	(0.0524)	(0.0642)
κ_X (exporter)	−0.221***	−0.230***	−0.106	−0.154***	−0.155**	−0.151
	(0.0548)	(0.0576)	(0.124)	(0.0593)	(0.0624)	(0.139)
Kleibergen–Paap F	419.65	360.27	71.11	419.48	360.17	70.99
p-value ($\kappa_M = \kappa_X$)	0.69	0.87	0.13	0.95	0.67	0.35
Panel B. controlling for productivity						
κ_M (importer)	−0.188***	−0.199***	−0.268***	−0.133***	−0.145***	−0.256***
	(0.0419)	(0.0472)	(0.0552)	(0.0413)	(0.0464)	(0.0557)
κ_X (exporter)	−0.215***	−0.219***	−0.0711	−0.142***	−0.133**	−0.0712
	(0.0532)	(0.0558)	(0.121)	(0.0527)	(0.0555)	(0.122)
β_ϕ (productivity)	−0.350***	−0.347***	−0.260***	−0.709***	−0.709***	−0.628***
	(0.00402)	(0.00414)	(0.00484)	(0.00420)	(0.00434)	(0.00560)
Kleibergen–Paap F	419.62	360.23	71.14	419.45	360.14	71.02
p-value ($\kappa_M = \kappa_X$)	0.66	0.76	0.07	0.89	0.85	0.09
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	393695	370509	202757	393787	370602	202853
Firms	67946	63457	45544	67951	63459	45548

Notes:
(i) This table shows the IV regression results of firm-level emission intensities on importer and exporter statuses. The regression specification is presented in Eq. (20). ζ^o and ζ^v denote the firm-level emission intensities normalized by the value of real total output and value-added, respectively. κ_M and κ_X are the coefficients of the dummy variables for the importer and exporter statuses respectively. β_ϕ is the coefficient of the natural logarithm of labor productivity. The instruments for the endogenous variables are discussed in Section 3.4.
(ii) The samples for this regression table do not include *dirty importers* defined in Section 4.4.2.
(iii) Panels A and B show the regression results without and with firm-level productivity in the regressors, respectively.
(iv) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

For the exporter coefficients in Table 7, the signs are mostly negative and significant, but exporter coefficients in columns (3) and (6) become insignificantly different from zero. One possible explanation is that the firms in the sub-sample tend to be in cleaner industries and those firms feature relatively low emission intensities regardless of their trade statuses. Thus, the causal effects of becoming an exporter may be more difficult to identify. Finally, the coefficients of the productivity term are consistently negative and significant, although we note that the inclusion of the productivity term does not alter the estimates of κs significantly. Overall, Table 7 shows that the negative effects of importing on firm-level emission intensities can only be partially explained by the POH.

4.5. Without accounting for importers

As we mentioned in introduction, studies have shown that exporting can effectively lower emission intensities at the firm level (Forslid et al., 2018; Shapiro, 2021; Rodrigue et al., 2022a), but the import-side effect received less attention.[55] This subsection illustrates that even when the researcher is only interested in the export side effect, it is important to include the firms' importer status as a covariate to estimate the accurate impact of exporting on emission intensities.

To illustrate this point, we replicate our baseline regression results in columns (1) through (3) of panel A of Table 5 without including the importer status and report the results in columns (1) through (3) of Table 8. Given that firms' importer and exporter statuses are generally positive correlated and the estimates of κ_D and κ_X are both negative, one might *a priori* expect the exclusion of the importer status would render the effect of exporter status even more negative. Compared with Table 5, however, we observe that the estimates for κ_X in Table 8 become less negative and even become positive in column (3).

We posit that this unexpected result is due to improper selection of the reference group. In specific, coefficients in Table 5 compare the conditional mean of exporting and exporting-and-importing firms' emission intensities with that of the reference group, which comprises of both importing-only and domestic firms. If importing-only firms account for a large share in the reference group, the gap of conditional means between the two groups narrows, which leads to a positive bias in the estimates of κ_X. In a severe case for *NOX*, importer status has a far greater negative impact than the exporter status (see Table 5) to the extent that the estimate of κ_X becomes positive.

[55] Recently, Cherniwchan (2017) studies the import channel of emission intensity reduction.

Table 8
Bias in the exporter coefficient without controlling for importer status.

Emis. Type	(1)	(2)	(3)	(4)	(5)	(6)
	All sample			No pure importers		
	COD	SO_2	NOX	COD	SO_2	NOX
κ_X (exporter)	−0.164***	−0.175***	0.207**	−0.257***	−0.345***	−0.646***
	(0.0492)	(0.0532)	(0.101)	(0.0600)	(0.0663)	(0.121)
Kleibergen–Paap F	634.27	565.00	123.81	499.70	420.24	113.57
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	422234	397146	215483	395602	375030	198862
Firms	69551	65488	47139	68102	64095	45315

Notes:
(i) This table reproduces the regression results in columns (1) through (3) in Panel A of Table 5 without including importer fixed effects. Regressions reported in columns (1) through (3) of this table use all firm samples, whereas those reported in columns (4) through (6) exclude firms that import but do not export products.
(ii) Standard errors robust against heteroskedasticity and clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To validate this explanation, we estimate additional regressions excluding importing-only firms and report the results in columns (4) through (6) of Table 8. Consistent with our interpretation, the estimates of κ_X become even more negative compared with our baseline results. This is due to removal of the reference group bias and the omitted-variable bias arising from the positive correlation between importer and exporter statuses that causes the exporter status term to absorb the overlapping effects.[56]

The results from this subsection suggest that there is a plausible and sizable bias in the exporter coefficients if we do not fully account for firms' full trading statuses. Thus, we strongly recommend future researchers to control for both importer and exporter statuses when the data are available.

4.6. Further robustness checks

We briefly describe several robustness checks of our main regression results.

4.6.1. Domestic vs. import/export-only firms

In this section, we aim to examine how the coefficients of the importing and exporting are identified in our regressions. For instance, is the coefficient of κ_M mainly identified by the difference between domestic firms and pure-importing firms or the difference between pure-exporting firms and two-way trading firms? Although these two differences are isomorphic in our theoretical framework, our data structure allows us to answer this question by strategically focusing on subsamples of firms. The results are presented in Appendix Table G.2. In columns (1) thorough (3) we exclude all exporting firms and simply compare domestic and importing firms; in columns (4) through (6), we exclude all importing firms and compare domestic and exporting firms. Compared with panel A of Table 5, we note that most coefficients still remain negative and significant. Yet, we find that some coefficients (e.g., κ_M for SO_2) become insignificant, whereas some (e.g., κ_X for COD) become greater in magnitude. The results in Appendix Table G.2 suggest that, in general, it is difficult to attribute to the identification of κs to either channel. Thus, we adhere to our main results in Table 5 in forming our conclusion, since they fully exploit the data and do not preempt any particular source of identification.

4.6.2. Processing vs. ordinary trade

A distinguishing feature of Chinese trade is prevalence of processing trade (Dai et al., 2016). Studies have shown that the consequences of processing trade can be different from ordinary trade.[57] To investigate whether the causal effect is entirely driven by processing trade, we focus on ordinary-trade exporters and importers. In specific, we note that the distribution of processing and ordinary trade is highly dichotomous for a single firm and thus we define *processing firms* if a firm's processing export share exceeds 90% in a single year. Next, we replicate our main regression results in Table 5 after excluding processing firms from the sample. The results are presented in Appendix Table G.3. Compared with Table 5, we note that the magnitude of all coefficients increase slightly, suggesting that ordinary trade may have a greater impact on firm-level emission intensities than processing trade. Yet, there is a caveat to this implication since, by definition, processing exporters engage in both importing and exporting, and thus it is perhaps difficult to decompose the causal effect of processing trade based on κ_M and κ_X. For the same reason, we cannot replicate Appendix Table G.3 in a way to compare processing firms and domestic firms.

[56] A minor caveat for interpreting the results in columns (4) through (6) in Table 8 is that the number of observations is approximately 7% less than that in our baseline results. As for the omitted-variable bias, one can overcome this issue by exerting instruments that are correlated with firms' exporter status but *not* with their importer status. To our knowledge, such IV is difficult to obtain.

[57] For example, Yu (2015) shows that processing-trade exporters are much less productive than ordinary-trade exporters.

4.6.3. Endogeneity associated with firm-level productivity

In our baseline regression results, we use a firm's labor productivity as the proxy for each firm's innate productivity. As pointed out by Olley and Pakes (1996), this approach is subject to the endogeneity problem in terms of estimating ϕ. To address this concern, we estimate firm-level productivity à la Ackerberg et al. (2015) (ACF) and re-estimate the regression model in panel B of Table 5. The results are presented in Appendix Table G.4. We note that both point estimates and statistical significance remain relatively the same, the only exception being κ_X for NOX, which becomes statistically insignificant. This can be explained by observing the sharp decline (around 10%) in the number of observations in Appendix Table G.4.

The sharp decline in the number of observation brings about two additional statistical challenges. First, a necessary condition to estimate the ACF productivity for a firm is that its real capital must be calculated with the perpetual inventory method. This suggests that the ACF productivity estimates for less productive firms (whose data are generally less complete) are missing. Naturally, this causes the "censoring from below" problem especially for domestic firms, making the comparison between domestic and trading firms more challenging.[58] Second, fewer observations naturally lead to more challenging statistical inference, especially given that κ_X for NOX is only significant at 10% level.

As we demonstrate in Table 5, the inclusion of the productivity term does not seem to be crucial to the κ estimates *conditional on* firm, province-year and industry-year fixed effects.[59] Thus, we prefer using labor productivity as our preferred specification mainly due to the benefit of exploiting the full dataset — with reassurance from Appendix Table G.4 that the results remain robust even if we use the ACF productivity as an alternative measure of firm productivity. If anything, Appendix Table G.4 suggests that the import channel, the focus of our study, is more robust than the export channel.

4.6.4. Product mix margin

Barrows and Ollivier (2018) identifies the product-mix margin, in which there can be a within-firm composition effect across different products with different levels of emission intensities. Our dataset does not allow us to calculate firm-product-specific emission intensities. However, one way to investigate whether the product-mix margin plays a significant role is to divide trading firms into two groups: the first group exporting 4 or fewer HS 4-digit products and the second group exporting more than 4 HS 4-digit products. We choose 4 products as the threshold since this is the median number of export products. Presumably, the product-mix margin would play a minimal role for the first group, and thus we compare domestic firms with two groups separately in order to investigate whether the negative impact of international trade on emissions is contingent on the product-mix margin.

The regression results are presented in Appendix Table G.5. In panel A, we present the regression results that compare domestic firms and trading firms whose number of export products is fewer than or equal to 4; in panel B, we compare domestic firms and trading firms whose number of export products is more than 4. There are two main takeways. First, in general, the negative impact of international trade on emissions is robust for both groups of trading firms. Second, we do not observe a monotonic relationship between the coefficient estimates in the two panels. In specific, κ_M for COD is greater in magnitude in panel A, but κ_X is greater in magnitude in panel B. In contrast, NOX emissions display an opposite pattern.

We view our simple exercise only as a preliminary attempt to understand the role of the product-mix margin and conclude that our main results do not depend on the product-mix margin (or lack thereof). We leave further investigation of the product-mix margin for future research.

5. Conclusion

Since its accession to the WTO in 2001, China's GDP and trade volume have increased rapidly. China has also become one of the top countries that contribute to various environmental emissions. Against this backdrop, it is crucial to understand how international trade has contributed to patterns of emissions in China.

This paper shows that both importing and exporting enhance firm-level abatement investment and reduce firm-level emission intensities. In our baseline IV regressions, we find that both importing and exporting lead to sizable reduction in emission intensities among Chinese firms. To validate the mechanism, we show that importing and exporting lead to a significant increase in firm-level water treatment capacities and removal of pollutants. Additionally, we demonstrate that our main regression results remain robust even after we account for the POH channel and researchers should control for firms' importer and exporter statuses whenever the data are available.

Our emphasis on firm-level adjustment mechanism enriches our understanding of how globalization impacts the environment, and we believe that insights from our paper can be extended to other developing countries that experience rapid trade liberalization.

[58] The sample selection bias here is distinct from the survival bias discussed in Olley and Pakes (1996). Specifically, the survival bias correction suggested in Olley and Pakes (1996) will lead to unbiased estimates of the productivity term, but it does not serve as a remedy for the bias in κs in our paper that arises from the sample selection bias.

[59] This resonates the results in Appendix Table G.1 which demonstrates that the fixed effects along with the labor productivity term account for over 62% variation in the ACF productivity.

Appendix A. Extension of the baseline model

A.1. Export decisions

Previous studies have shown that importing firms also tend to be exporters (Amiti et al., 2014; Blaum, 2019). Thus we extend our baseline model to allow a subset of importing firms to export their products to the foreign country. Without loss of generality, we consider a special case in which the exporter cutoff productivity is strictly higher than the importer cutoff. Thus, all exporting firms are importers but not vice versa. In Appendix C, we describe how a further extension of our baseline model can account for any trading statuses (i.e., exporting but not importing).

Let $\pi^X(\phi)$ denote the profit of a firm that exports (and imports). Based on the previous discussion, we can express an exporting firm's profit as

$$\pi^X(\phi) = \frac{1-\rho(\sigma-1)}{\rho(\sigma-1)}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\left(EP^{\sigma-1} + E^*\left(P^*/\delta\right)^{\sigma-1}\right)\tilde{\mu}^{*1-\sigma}\phi^{\sigma-1}\rho\right]^{\frac{1}{1-\rho(\sigma-1)}} - f_I - f_X. \tag{A.1}$$

where f_X is the fixed cost of exporting and δ is the iceberg transportation cost.

An importing firm will also export if

$$\pi^X(\phi) > \pi^M(\phi).$$

Analogous to Proposition 3, we can characterize the export decisions using a single productivity cutoff ϕ^{**}, above which importing firms become exporters.

Proposition A.1. *More productive importing firms become exporters. Specifically, firms become exporters if their productivity ϕ is greater than ϕ^{**}, where ϕ^{**} is given by*

$$\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\tilde{\mu}^{*1-\sigma}\phi^{**\sigma-1}\rho\right]^{\frac{1}{1-\rho(\sigma-1)}}\left\{\left[EP^{\sigma-1} + E^*(P^*/\delta)^{\sigma-1}\right]^{\frac{1}{1-\rho(\sigma-1)}} - \left(EP^{\sigma-1}\right)^{\frac{1}{1-\rho(\sigma-1)}}\right\} = f_X\frac{\rho(\sigma-1)}{1-\rho(\sigma-1)}. \tag{A.2}$$

Proposition A.1 shows that firms stratify themselves into different trading patterns based on their productivity levels. The least productive firms are domestic, more productive firms import intermediate inputs, and the most productive firms choose to both import intermediate inputs and export their final products to the foreign country. In general, the cutoff productivity ϕ^* for importing can be above that for exporting ϕ^{**}. However, for the brevity of discussion we focus on a special case that $\phi^* < \phi^{**}$, because discussing the alternative case does not yield additional insights.

Assumption 1. *In the baseline model, we consider the case in which the fixed cost of exporting is sufficiently high such that $\phi^{**} > \phi^*$ in equilibrium. Namely, the cutoff productivity for a firm to be an exporter is higher than that to be an importer.*

Assumption 1 is for the brevity of analysis and this can be easily relaxed as we show in Appendix C. For Assumption 1 to be satisfied in equilibrium, it is required that the fixed cost of exporting is sufficiently higher than that of importing.

The next proposition shows that holding constant the level of firm productivity, a firm's decision to export final goods leads to an increase in abatement investment and hence a reduction in emission intensity.

Proposition A.2. *For a given level of firm productivity ϕ, an exporting-and-importing firm features a higher level of abatement investment and a lower level of emission intensity compared with what they would be if the firm were importing and not exporting. Formally, we have*

$$\begin{aligned} &f_A^X(\phi) > f_A^M(\phi), \\ &\zeta^X(\phi) < \zeta^M(\phi). \end{aligned} \tag{A.3}$$

Proof. Conditional on exporting, the level of abatement investment of a firm with productivity level ϕ is given by

$$\left[f_A^X(\phi)\right]^{1-\rho(\sigma-1)} = \left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\left(EP^{\sigma-1} + E^*(P^*/\delta)^{\sigma-1}\right)\tilde{\mu}^{*1-\sigma}\phi^{\sigma-1}\rho,$$

which is greater than $f_A^M(\phi)$ shown in Eq. (13). Similarly, the emission intensity of an exporting firm can be expressed as

$$\zeta^X(\phi) = \frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\left(EP^{\sigma-1} + E^*(P^*/\delta)^{\sigma-1}\right)\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}^*}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}}, \tag{A.4}$$

which is lower than $\zeta^M(\phi)$ presented in Eq. (18).

The findings in this subsection are also summarized in Fig. 2. The figure shows that around the exporter cutoff ϕ^{**}, there is a continuous increase in profit, and yet there are discontinuities in output, abatement investment and emission intensities. Fig. 2 clearly shows that exporting lowers firms' emission intensities.

In our empirical analysis, we test Propositions 4 and A.2 with Chinese firm-level panel data on emissions. Based on our theory, we conjecture that firms should exhibit lower emission intensities if they engage in international trade, even after we account for the differences in productivity.

Table A.1
Model parameters and calibration.

Parameters	Value	Moment	data	Model
$\tilde{E}^*$	1.330	share of export revenue for exporting firms	0.248	0.248
f_I	0.027	share of importing firms	0.229	0.229
f_X	0.120	share of importing and exporting firms	0.143	0.143
γ	0.284	share of output accounted for by importing firms	0.408	0.408
υ_m	0.930	cost share of intermediate input	0.930	0.930
υ_f	0.665	cost share of imported intermediate input	0.195	0.195

Notes: This table shows the parameter details of the calibrated model. See text in Appendix A.2 for details.

A.2. Model calibration

In this section, we calibrate our model using the key data moments. The purpose is twofold. First, we demonstrate the robustness of our theoretical prediction under a more realistic set of parameter values and second, we perform basic counterfactuals to understand the relevance of the import channel.

To facilitate calibration, we first consolidate model parameters by introducing $\tilde{E}$ and $\tilde{E}^*$ to denote $\tilde{E} \equiv EP^{\sigma-1}$ and $\tilde{E}^* \equiv E^*(P^*/\delta)^{\sigma-1}$. We normalize parameters $\{\tilde{E}, c_d, c_f, w, \tau, \upsilon_d\}$, which are meaningful only in their values relative to other parameters, to be 1. Next, without compromising the qualitative property of our calibrated model, we either adopt the parameter values widely used in the literature ($\sigma = 3$ (Caliendo and Parro, 2015), $\alpha = 0.03$ (Shapiro and Walker, 2018)) or assume values that do not conflict with model assumptions ($\eta = 4$ and $\rho = 0.2$).[60]

The remaining parameters are calibrated to match the relevant empirical moments and are presented in Table A.1. The model is exactly identified with six parameters and six corresponding empirical moments. Next, we discuss how the model parameters govern the key relevant empirical moments. The share of export revenue for exporting firms is determined by $\tilde{E}^*$ according to Eq. (A.1). f_I and f_X determine the importing and exporting productivity cutoffs, respectively, and thus we can rely on their values to target the shares of importing and exporting firms in the data. Following Helpman et al. (2004), we assume that the firm productivity ϕ follows a Pareto distribution with a scale parameter $\underline{\phi} = 2$ (without loss of generality) and a shape parameter $\gamma > 0$. Intuitively, the shape parameter governs the relative size of trading firms versus domestic firms as a greater shape parameter shrinks the right tail of the distribution. Parameters $\{\upsilon_m, \upsilon_f\}$ jointly determine the cost shares of total intermediate inputs and imported intermediate inputs. The final remaining parameter υ_l is determined by the relation $\upsilon_l = 1 - \upsilon_m$.

The model equilibrium is shown numerically in Fig. A.1. In panel (a), we plot the profit of firms over the productivity distribution under the scenarios that they choose not to trade (purple circles), import intermediate inputs (blue line) and export final goods in addition to importing (red dashed line). As we demonstrate in Proposition A.1, there are clear cutoffs determined by ϕ^* and ϕ^{**}, above which firms change their trading status. In panels (b) and (c) of Fig. A.1, we plot firms' abatement investment and emission intensities under alternative trading statuses. The figures confirm our Proposition A.2 that the exporting decision increases firms' abatement investments and reduces their emission intensities.

The calibrated model allows us to evaluate how import-side trade liberalization affects firm-level abatement investments and emission intensities. Specifically, we let the price of the imported input c_f drop by 20% and examine how firms adjust their abatement investment. In the counterfactual analysis, we find that importing firms jointly increase their abatement investment by 16.1% and their average emission intensities drop by 7.2%. The verdict on the direction of total emissions, however, is complicated by two forces: (i) due to import-side trade liberalization, importing firms can increase their production and (ii) importing firms invest more in abatement to reduce their emission intensities. On balance, our counterfactual results suggest that total emissions in the economy increase by 0.33% as a result of import-side trade liberalization.

A.3. Impact of trade liberalization on domestic firms

We have shown that both import- and export-side trade liberalizations lower emission intensities of firms that engage in international trade. This naturally leads to two important questions. What is the impact of trade liberalization on domestic firms? What is its impact on economy-wide total emissions? We answer these two questions in the remainder of the theoretic section.

To this end, we introduce two additional equations that endogenize the equilibrium measure of firms $\mathcal{M}$ and the price index in the home country P. The equilibrium measure of firms is pinned down by a free entry condition:[61]

$$f_E = \int_{\underline{\phi}}^{\phi^*} \pi^D(\phi) dG(\phi) + \int_{\phi^*}^{\phi^{**}} \pi^M(\phi) dG(\phi) + \int_{\phi^{**}}^{\infty} \pi^X(\phi) dG(\phi), \tag{A.5}$$

[60] As a robustness check, we allow η and ρ to vary within reasonable ranges $(3, 5)$ and $(0.15, 0.25)$ respectively. Both quantitative and qualitative model properties remain highly consistent.

[61] In contrast to the Melitz (2003) model, we do not have a fixed cost of operating in the domestic market. Thus, we abstract from firms that pay the entry cost f_E but find it unprofitable to operate in the domestic market. The heterogeneous fixed cost model that we present in Appendix C can account for firms that either import or export but do not have any domestic sales if we further assume a fixed cost for domestic operation.

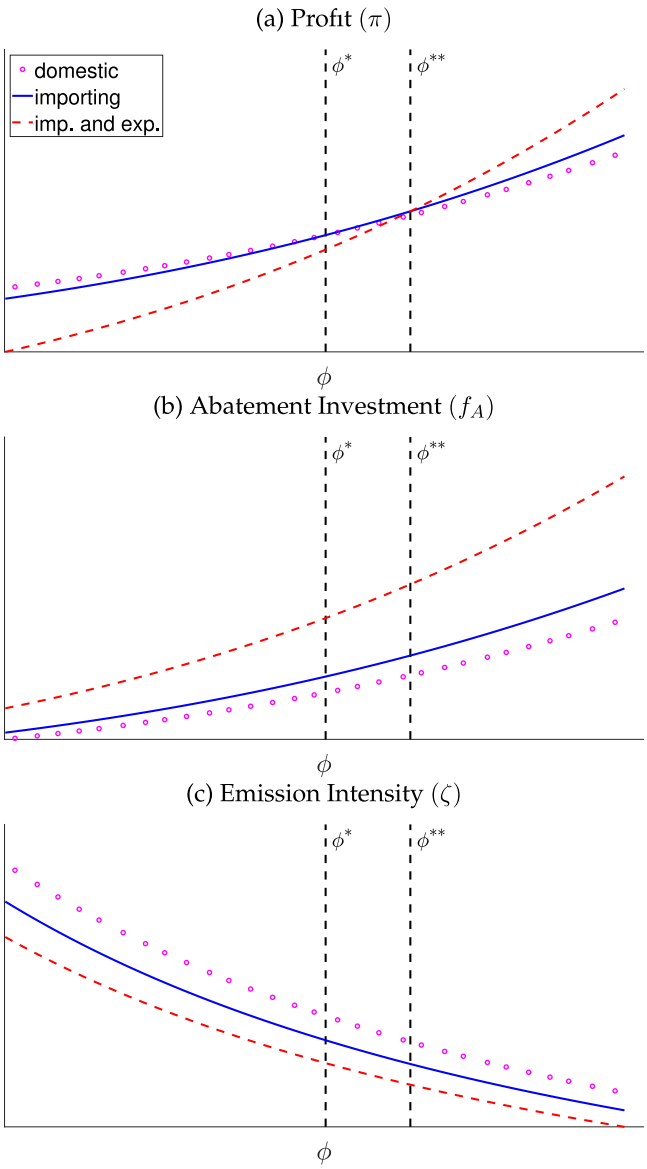

Fig. A.1. Model equilibrium with calibrated parameters.

where f_E is the sunk cost of entry, $G(\phi)$ is the c.d.f. of the productivity distribution of firms, and $\underline{\phi}$ is the lower bound of the productivity distribution. To understand Eq. (A.5), consider a potential entrant to the market that pays f_E to draw a level of productivity ϕ from the distribution $G(\phi)$. Depending on the level of productivity, the firm decides whether to import intermediate inputs and whether to export final goods. The expected operating profit is given by the right-hand side of Eq. (A.5). The expected profit decreases with the measure of operating firms until the equilibrium in Eq. (A.5) renders firm entry indifferent.

The second equation for determining the economy-wide equilibrium is the explicit expression for the price index in the home country:

$$P^{1-\sigma} = \mathcal{M}\left[\int_{\underline{\phi}}^{\phi^*} p^D(\phi)^{1-\sigma} dG(\phi) + \int_{\phi^*}^{\phi^{**}} p^M(\phi)^{1-\sigma} dG(\phi) + \int_{\phi^{**}}^{\infty} p^X(\phi)^{1-\sigma} dG(\phi)\right] + P_f^{1-\sigma}, \tag{A.6}$$

where P_f is the average price index of foreign final goods sold in the home country.[62]

[62] The measure of foreign firms is implicitly defined in P_f; thus, in Eq. (A.6) we do not multiply the price index of foreign goods by the measure of foreign firms.

With equations Eq. (A.5) and Eq. (A.6), the conventional intra-industry reallocation is at work against domestic firms à la Melitz (2003). Specifically, when trade liberalization benefits trading firms in the home country, domestic firms experience a negative demand shock as it becomes more difficult to compete with trading firms that reduce their marginal costs via using foreign intermediate inputs and increasing investment in abatement.[63] A negative demand shock, in turn, *reduces* domestic firms' abatement investment and increases their emission intensity. We summarize this result in the following proposition.

Proposition A.3. *Both import- and export-side trade liberalizations reduce domestic firms' abatement investment and increase their pollution intensity.*

Proof.

Consider import-side trade liberalization such that $\tilde{\mu}^{*\prime} < \tilde{\mu}^{*}$, where we use an extra superscript "$\prime$" to denote the equilibrium level of variables after trade liberalization. As we have characterized the equilibrium levels of abatement investment and emission intensity in equations Eq. (13) and Eq. (17), it suffices to show that $P' < P$ to prove Proposition A.3. We prove this by contradiction. Suppose hypothetically that $P' > P$.

In the new equilibrium, a lower marginal cost $\tilde{\mu}^{*\prime} < \tilde{\mu}^{*}$ induced by trade liberalization means that both productivity cutoffs for importing and exporting decrease:

$$\begin{aligned} \phi^{*\prime} &< \phi^{*}, \\ \phi^{**\prime} &< \phi^{**}. \end{aligned} \tag{A.7}$$

Then, it is possible to show that firms that used to import and export experience increases in their profits:

$$\int_{\phi^{*}}^{\phi^{**}} \pi^{M}(\phi)dG(\phi) + \int_{\phi^{**}}^{\infty} \pi^{X}(\phi)dG(\phi) < \int_{\phi^{*}}^{\phi^{**\prime}} \pi^{M\prime}(\phi)dG(\phi) + \int_{\phi^{**\prime}}^{\infty} \pi^{X\prime}(\phi)dG(\phi). \tag{A.8}$$

Additionally, some domestic firms become importing firms and experience increases in their profits:

$$\int_{\phi^{*\prime}}^{\phi^{*}} \pi^{D}(\phi)dG(\phi) < \int_{\phi^{*\prime}}^{\phi^{*}} \pi^{M\prime}(\phi)dG(\phi). \tag{A.9}$$

Finally, the profits of domestic firms increase as well since $P' > P$.

$$\int_{\underline{\phi}}^{\phi^{*\prime}} \pi^{D}(\phi)dG(\phi) < \int_{\underline{\phi}}^{\phi^{*\prime}} \pi^{M\prime}(\phi)dG(\phi). \tag{A.10}$$

Summarizing equations (A.8), (A.9) and (A.10), it is easy to see that the free entry condition in Eq. (A.5) is violated as the profits of firms of all types increase. Thus, we have shown by contradiction that $P' < P$.

From equations (13) and (17), it is clear that as the price index declines as a result of trade liberalization, domestic firms endogenously choose lower abatement investment and higher emission intensity.

For export-side trade liberalization, the proof is similar. It is easy to show that if it becomes easier to export final goods, exporting firms' profits increase and the economy-wide price index has to decrease as a result. Then, by a similar force, domestic firms' abatement investment and emission intensity fall.

As we have shown in Propositions 4 and A.2, trade liberalization allows more firms to become importers and exporters and potentially contributes to lowering economy-wide emission intensity. Proposition A.3, however, shows that this logic is complicated by domestic firms that increase their pollution intensity. Accordingly, we explore the implications of trade liberalization for economy-wide emissions in the next section.

A.4. Impact of trade liberalization on total emissions

To explore the implications for the total emissions in the home country, we impose a balanced trade condition to close the model. As it turns out, depending on the type of the balanced trade condition, the model has different implications on the level of total emissions. Hence, we present two alternative ways to close the model and discuss their mechanisms.

The first one is the balanced trade condition imposed only on the final goods. This implies that the total value of final goods exports from the home country is equal to that of imports from the foreign country. Formally,

$$\underbrace{EP^{\sigma-1}P_f^{1-\sigma}}_{\text{final goods import}} = \underbrace{\int_{\phi^{**}}^{\infty} E^{*}\left(P^{*}/\delta\right)^{\sigma-1} p^{X}(\phi)^{1-\sigma} \cdot \mathcal{M}dG(\phi)}_{\text{total export of final goods}}. \tag{A.11}$$

We refer to this as the *balanced trade condition of final goods*. Under this balanced trade condition, imported intermediate inputs are paid to the foreign suppliers with the corresponding amount of numéraire good q_0.

[63] We define *trading firms* as importing firms and exporting firms.

An alternative way to impose the balanced trade condition is to let the total amount of home country's exports equal the sum of intermediate goods imports and the final goods import from the foreign country. Formally,

$$\underbrace{\left[\int_{\phi^*}^{\phi^{**}} EP^{\sigma-1}p^M(\phi)^{1-\sigma}\cdot\mathcal{M}dG+\int_{\phi^{**}}^{\infty}\left(EP^{\sigma-1}+E^*\left(P^*/\delta\right)^{\sigma-1}\right)p^X(\phi)^{1-\sigma}\cdot\mathcal{M}dG\right]\frac{\sigma-1}{\sigma}\tilde{v}_f}_{\text{intermediate goods import}}$$
$$+\underbrace{EP^{\sigma-1}P_f^{1-\sigma}}_{\text{final goods import}}=\underbrace{\int_{\phi^{**}}^{\infty}E^*\left(P^*/\delta\right)^{\sigma-1}p^X(\phi)^{1-\sigma}\cdot\mathcal{M}dG(\phi)}_{\text{total export of final goods}}, \tag{A.12}$$

where we use $\tilde{v}_f$ to denote the share of foreign inputs in the composite input bundle of production.[64] Under this condition, there are no cross-border transactions involving numéraire goods. We simply refer to this case as the *combined balanced trade condition*.[65]

To characterize the economy-wide emissions under two different balanced trade conditions, we establish the following lemma that characterizes the economy-wide pollution Z as a linear function of the total revenue in the home country.

Lemma A.1. *The total emissions in the home country are a linear function of its total revenue of final goods. Specifically,*

$$Z\equiv\int z(\phi)\cdot\mathcal{M}dG(\phi)=\frac{\sigma-1}{\sigma}\frac{\alpha v_l}{\tau}R, \tag{A.13}$$

where R is the economy-wide revenue of the final goods, defined as

$$R\equiv\underbrace{\int_{\underline{\phi}}^{\phi^*}EP^{\sigma-1}p^D(\phi)^{1-\sigma}\cdot\mathcal{M}dG(\phi)}_{\textit{total revenue of domestic firms}}+\underbrace{\int_{\phi^*}^{\phi^{**}}EP^{\sigma-1}p^M(\phi)^{1-\sigma}\cdot\mathcal{M}dG(\phi)}_{\textit{total revenue of importing firms}}$$
$$+\underbrace{\int_{\phi^{**}}^{\infty}\left[EP^{\sigma-1}+E^*\left(P^*/\delta\right)^{\sigma-1}\right]p^X(\phi)^{1-\sigma}\cdot\mathcal{M}dG(\phi)}_{\textit{total revenue of exporting firms}}. \tag{A.14}$$

Proof. Let $z(\phi|t)$ denote the firm-level emissions of a firm with productivity level ϕ and trading type t. The previous discussion shows that

$$z(\phi|t)=\frac{\alpha v_l(\sigma-1)}{\tau\sigma}EP^{\sigma-1}p(\phi|t)^{1-\sigma}\qquad\text{for } t=D,M,$$
$$z(\phi|t)=\frac{\alpha v_l(\sigma-1)}{\tau\sigma}\left(EP^{\sigma-1}+E^*\left(P^*/\delta\right)^{\sigma-1}\right)p^t(\phi)^{1-\sigma}\qquad\text{for } t=X.$$

Thus, the economy-wide emissions Z can be expressed as

$$\begin{aligned}Z&=\int z(\phi)\cdot\mathcal{M}dG(\phi)\\&=\frac{\alpha v_l(\sigma-1)}{\tau\sigma}\mathcal{M}\left[\int_{\underline{\phi}}^{\phi^*}EP^{\sigma-1}p^D(\phi)^{1-\sigma}dG(\phi)+\int_{\phi^*}^{\phi^{**}}EP^{\sigma-1}p^M(\phi)^{1-\sigma}dG(\phi)\right.\\&\quad\left.+\int_{\phi^{**}}^{\infty}\left(EP^{\sigma-1}+E^*\left(P^*/\delta\right)^{\sigma-1}\right)p^X(\phi)^{1-\sigma}dG(\phi)\right]\\&=\frac{\alpha v_l}{\tau}\frac{\sigma-1}{\sigma}R\end{aligned}.$$

It is noteworthy that Lemma A.1 holds regardless of which balanced condition is imposed. Moreover, Lemma A.1 is very useful in the sense that to characterize the economy-wide emissions in the home country, we only need to characterize the total revenue of final goods in the home country and do not need to worry about the firm-level heterogeneity in abatement investment.

The following proposition shows that under the balanced trade condition of final goods, the total emissions in the home country are always a constant.

Proposition A.4. *Under the balanced trade condition of final goods, the economy-wide pollution is neutral to trade liberalization. Specifically, the total emissions in the home country are given by*

$$Z=\frac{\alpha v_l}{\tau}\frac{\sigma-1}{\sigma}E, \tag{A.15}$$

where E is the home consumers' expenditure on final goods specified in the utility function in Eq. (1).

[64] $\tilde{v}_f\equiv v_m\frac{v_f^{\eta}c_f^{1-\eta}}{v_d^{\eta}c_d^{1-\eta}+v_f^{\eta}c_f^{1-\eta}}$

[65] Note here that the definitions "intermediate" and "final" are from the home country's perspective. In other words, the home country's exports of final goods can be used as intermediate inputs for foreign firms. However, the foreign country's usage of the home country's exports does not play a role in our analysis.

Proof. The balanced trade condition of final goods suggests that

$$\underbrace{\int_{\phi^{**}}^{\infty} E^* \left(P^*/\delta\right)^{\sigma-1} p^X(\phi)^{1-\sigma} \cdot \mathcal{M} dG(\phi)}_{\text{home final goods export}} = \underbrace{EP^{\sigma-1} P_f^{1-\sigma}}_{\text{final goods import}}. \tag{A.16}$$

Multiplying both sides of Eq. (A.6) by $EP^{\sigma-1}$, we obtain the expression for the final goods expenditure in the home country:

$$\begin{aligned} E =& \mathcal{M}\left[\int_{\underline{\phi}}^{\phi^*} EP^{\sigma-1} p^D(\phi)^{1-\sigma} dG(\phi) + \int_{\phi^*}^{\phi^{**}} EP^{\sigma-1} p^M(\phi)^{1-\sigma} dG(\phi)\right. \\ &\left. + \int_{\phi^{**}}^{\infty} EP^{\sigma-1} p^X(\phi)^{1-\sigma} dG(\phi)\right] + EP^{\sigma-1} P_f^{1-\sigma}. \end{aligned} \tag{A.17}$$

After equation (A.16) is plugged into equation (A.14), it immediately becomes clear that $R = E$. Then, by Lemma A.1, we show that the total emissions in the home country is given by the expression in equation (A.15).

Proposition A.4 shows that although trade liberalization causes economy-wide reallocation in favor of importing and exporting firms, the total emissions remain constant as long as the total expenditures on final goods imports and exports are balanced. Some readers might wonder if this result critically hinges on our assumption that the total expenditure on the final goods is exogenous and given by "E" in our model. Forslid et al. (2018) consider a model in which expenditure is determined endogenously by the national income. They also show that trade liberalization has no impact on the economy-wide total emissions. Therefore, to interpret our results in the context of the literature, we extend the result of Forslid et al. (2018) to show that the emission neutrality holds in broader cases.[66]

The key difference of our result from that of Forslid et al. (2018), however, is that Proposition A.4 hinges less on the country *symmetry* assumption that is critical for the result of Forslid et al. (2018). Forslid et al. (2018) assume that home and foreign countries are symmetric, and therefore the export price index (inclusive of iceberg trade costs) is identical to the import price index. In our model, we do not impose a symmetry assumption on home and foreign countries. In other words, Proposition A.4 allows for an arbitrary asymmetry between home and foreign countries in terms of country sizes, trade costs and productivity distributions.[67]

In what follows, we discuss our next proposition regarding the equilibrium total emissions in the home country under the combined balanced trade condition. As it turns out, import-side trade liberalization increases the total emissions in the home country in this case.

Proposition A.5. *Under the combined balanced trade condition, the total emissions in the home country are increasing linearly with the total intermediate goods imports. Specifically,*

$$Z = \frac{\sigma-1}{\sigma}\frac{\alpha v_l}{\tau}\left[E + IN\right], \tag{A.18}$$

where IN refers to the volume of total intermediate goods imports defined as

$$IN = \left[\int_{\phi^*}^{\phi^{**}} EP^{\sigma-1} p^M(\phi)^{1-\sigma} \cdot \mathcal{M} dG + \int_{\phi^{**}}^{\infty} \left(EP^{\sigma-1} + E^*\left(P^*/\delta\right)^{\sigma-1}\right) p^X(\phi)^{1-\sigma} \cdot \mathcal{M} dG\right] \frac{\sigma-1}{\sigma}\tilde{v}_f.$$

Proof. Under the combined balanced trade condition, the total revenue in the home country R can be written as

$$\begin{aligned} R \equiv& \int_{\underline{\phi}}^{\phi^*} EP^{\sigma-1} p^D(\phi)^{1-\sigma} \cdot \mathcal{M} dG(\phi) + \int_{\phi^*}^{\phi^{**}} EP^{\sigma-1} p^M(\phi)^{1-\sigma} \cdot \mathcal{M} dG(\phi) \\ &+ \int_{\phi^{**}}^{\infty} EP^{\sigma-1} p^X(\phi)^{1-\sigma} \cdot \mathcal{M} dG(\phi) + EP^{\sigma-1} P_f^{1-\sigma} + IN \\ =& E + IN. \end{aligned}$$

By Lemma A.1, Eq. (A.18) is proven.

Proposition A.5 provides a new insight into the impact of import-side trade liberalization on emissions. Namely, when such trade liberalization leads to an expansion of intermediate input imports, the total output in the home country needs to increase by the same amount to restore the combined balanced trade condition.

Some might wonder, whether a reduction in final goods imports can offset the increase in intermediate goods imports so that the left-hand side of Eq. (A.12) stays constant. It turns out that the total emissions in the home country will still increase in this case

[66] One potential way to generate nonneutrality of trade liberalization for emissions is to consider multi-sector final goods such that sectors differ in the underlying parameters governing the pollution intensity. Trade liberalization causes a rescaling of different sectors and hence there can be either a negative or a positive impact on the total emissions. However, this channel is not central to our research question, and we leave this to future research.

[67] However, we remind readers that a direct comparison between our results and the results of Forslid et al. (2018) may be difficult as we assume that the representative consumer's utility function takes a quasi-linear form. This assumption provides the analytic convenience of muting the general equilibrium effects whereby the individual firms interact with each other through the determination of factor prices and the aggregate income.

because the decrease in final goods imports will be offset by the same amount of increment in output sold domestically as shown in Eq. (A.17). In Eq. (A.17), the sum of final goods imports and the sales of home firms in the home country is a constant given by E.

A.5. Discussion of model assumptions

In this section, we discuss a few model assumptions that ease the exposition of results. These assumptions are not central to our results, and one can easily relate to alternative cases in which the assumptions are violated.

A.5.1. Assumption that the export cutoff productivity is higher than the import cutoff productivity

In our baseline model, we assume that exporting requires a higher productivity than importing $\phi^* < \phi^{**}$. However, this assumption is not crucial to our results. If we consider the opposite case of $\phi^* > \phi^{**}$ most of our results will remain the same.

In Appendix C, we present an extension of our model in which the fixed costs of importing and exporting are firm-specific. In this case, firms' decisions to import and export cannot be characterized by single cutoffs. Instead, they are determined by each individual firm's productivity relative to its own fixed cost of importing and exporting. The purpose of this extension is to account for the empirical observation that firms' import and export decisions are not perfectly aligned with firms' productivity. As shown in Appendix C, the key results that we derive from our baseline model still hold under this extension.

A.5.2. Assumption on trade balance

In our baseline model, we close the model by assuming that trade is balanced. We do so since we believe that this condition offers a clear starting point for thinking about the economy-wide total emissions. In light of the massive trade surplus in China, one can easily relate to a further model extension in which trade is not balanced. If the home country experiences a trade surplus, it means that the total revenue increases by the corresponding amount and, by Lemma A.1, it is easy to predict that the total emissions in the home country increase. Thus, through the lens of our model, it can be argued that if international trade increases emissions in China (or in developing countries in general), it is mainly driven by trade surplus.

Appendix B. Procedure for constructing the merged dataset

B.1. Cesd-asie matching

CESD and ASIE have two major firm identifiers: firm name and organization codes. We also use location as a secondary identifier. Firm location is stored in different variables in the two datasets. In CESD, firm location is indicated by the six-digit administrative division code. The first two digits corresponds to province-level administrative division (the first-tier), the middle two digits corresponds to prefecture-level administrative division (the second-tier) and the last two digits corresponds to a county-level administrative division (the third-tier). In ASIE, firm location is indicated by the zip code. The first two digits of a zip code correspond to the province, the first three identify the mailing district, and the first four identify prefecture/county. Since administrative division codes and zip codes do not have a strict one-to-one mapping, we identify matched administrative division code and zip code pairs to improve matching quality.

We first perform exact matching on name and organization code that are uniquely identified in both datasets. Next, we use the "MATCHIT" feature in Stata to create a similarity score for name pairs of unmatched firms in the remaining sample. The score measures the "generalized edit distance" of a pair of strings. Its value ranges from 0 to 1, with 1 signifying that the two strings are identical. We visually check the performance of fuzzy matching and set the similarity score to 0.7, the value that minimizes both false positives and false negatives. The matching procedure is as follows:

1. Merge the two datasets by exact firm name matches.
2. Merge the unmatched data from Step 1 by organization code.
3. Identify matched administrative division code and zip code pairs based on the matched data from Step 1. Perform fuzzy name matching between firms in matched districts.
4. Merge the unmatched data from Step 2 by matched fuzzy name pairs.
5. Append the matched data from Steps 1, 2 and 4. This generates the matched firm pairs between CESD and ASIE.

As shown by the Table B.1, approximately 50% firms have matched firms in ASIE. Fuzzy name matching increases the matching ratio by 10%. These firms are generally bigger than the unmatched firms since the matched firms account for approximately 75% of the total output in CESD.

B.2. Asie-cust matching

The procedure used to match the ASIE and CUST is similar to the above. We first perform exact matching of names and telephone numbers. Next, we perform fuzzy name matching for the unmatched sample. Our matching algorithm improves over Yu (2015), as we include fuzzy name matching that increases the matching ratio by more than 10% in early years and the overall matching ratio by 8%. As shown by Table B.2, our method results in the matching rate of 20% for firms in ASIE. These firms are larger than the unmatched firms, as their output ratio over the whole sample is around 35%.

Table B.1
CESD-ASIE matching.

Year	CESD observations	Match ratio	*Match ratio by steps*			Output ratio
			Exact name	Telephone number	Fuzzy name	
2000	70223	51%	28%	16%	7%	74%
2001	70187	51%	29%	15%	6%	73%
2002	68191	53%	31%	15%	6%	74%
2003	69323	52%	31%	16%	6%	77%
2004	70457	60%	39%	15%	6%	79%
2005	70093	60%	40%	15%	6%	79%
2006	76501	58%	39%	14%	5%	75%
2007	104058	55%	37%	13%	5%	79%
2008	108598	57%	41%	12%	5%	76%
2009	109149	47%	34%	5%	8%	65%
2010	111124	50%	33%	12%	4%	70%
2011	153027	44%	35%	6%	3%	80%
2012	147996	45%	36%	6%	3%	74%
2013	147657	47%	38%	6%	4%	76%

Table B.2
ASIE-CUST matching.

Year	ASIE observations	Match ratio	*Match ratio by steps*			Output ratio
			Exact name	Telephone number	Fuzzy name	
2000	162885	16%	12%	2%	2%	38%
2001	171256	17%	14%	2%	2%	40%
2002	181557	18%	15%	2%	2%	42%
2003	196222	19%	16%	1%	2%	42%
2004	279092	21%	18%	1%	2%	
2005	271835	20%	17%	2%	2%	41%
2006	301961	17%	14%	2%	1%	29%
2007	336768	20%	18%	1%	2%	38%
2008	411407	19%	17%	1%	1%	35%
2009	320778	20%	18%	1%	1%	34%
2010	348536	17%	15%	1%	1%	35%
2011	302593	20%	19%	0%	1%	31%
2012	311314	19%	19%	0%	0%	25%
2013	344875	21%	20%	0%	0%	33%

B.3. Panel construction

In the last step, we match CESD firms over time for potential panel analysis. The task of matching over time is similar to that of matching across datasets. There are two main identifiers in CESD: firm name and organization code. Eiether may change due to misreporting or a change of the reporting standard. A company name may correspond to multiple organization codes, and an organization code may correspond to multiple company names. We perform fuzzy name matching over time to improve the efficiency of matching by organization code. The matching method is as follows:

1. Find the firms with one-to-one matches of name and organization code.
2. For firms with a unique name but multiple organization codes, identify the same firm over time by considering location.
3. For a unique organization code with multiple company names, identify the same firm over time by considering similarity of firm names, continuity of records, location and establishment year.
4. For firm names with multiple organization codes and organization codes with multiple firm names, first identify the same firms under the same organization code, as in Step 3, and then identify the same firm with different organization code, as in Step 2.
5. Combine the four types of firms together. An ID is assigned to a firm and its identified matching firms through time.

Appendix C. Rationalizing the non-stratified productivity distribution of importing and exporting firms with the heterogeneous fixed cost extension

The first challenge in confronting our model with data is that firms' trading statuses are not strictly aligned according to their productivity. In our baseline model, all firms face the same importing and exporting costs, and thus firms self-select into importing and exporting statuses strictly based on their innate productivity ϕ. This is a convenient assumption for explaining why importing and exporting firms are more productive than domestic firms in general. However, this assumption is at odds with the common data pattern that some importing or exporting firms are not as productive as domestic firms.

To account for the non-stratified productivity distribution of importing and exporting firms, we relax the assumption on the homogeneous fixed costs of importing and exporting. We also show that our key testable implications of our model – Propositions 4 and A.2 – still hold under this extension.

Specifically, we suppose that ax-ante homogeneous firms incur a sunk cost of entry f_E and draw a firm-specific triplet ξ, defined as $\xi \equiv \{\phi, f_I, f_X\}$, from a joint distribution. In particular, ξ consists of inherent productivity ϕ, firm-specific fixed costs of importing f_I and firm-specific fixed costs of exporting f_X. Firms' importing and exporting decisions are determined by the two-dimensional extensive margin decisions on importing and exporting. We label each group as follows:

$$\begin{aligned}
\mathbb{D} &\equiv \{\xi \mid \operatorname*{argmax}_{i\in T} \pi(t,\xi) = D\},\\
\mathbb{M} &\equiv \{\xi \mid \operatorname*{argmax}_{i\in T} \pi(t,\xi) = M\},\\
\mathbb{X} &\equiv \{\xi \mid \operatorname*{argmax}_{i\in T} \pi(t,\xi) = X\},\\
\mathbb{Y} &\equiv \{\xi \mid \operatorname*{argmax}_{i\in T} \pi(t,\xi) = MX\},
\end{aligned} \tag{C.1}$$

where T is the set of trading types that consists of D, M, X and MX that, respectively, stand for being a domestic firm, an importer, an exporter and engaging in both importing and exporting. The term $\pi(i,\xi)$ is the profit of the firm associated with being type i with parameter ξ. Expressions in Eq. (C.1) mean that firms optimally choose their trading types to maximize the profit. $\mathbb{D}, \mathbb{M}, \mathbb{X}$ and $\mathbb{Y}$ denote, respectively, the set of firms that are domestic, importers, exporters, and both.

This small modification of our baseline model allows us to rationalize: (1) the coexistence of four types of firms in the data, and (2) the empirical pattern that firms' trading statuses are not perfectly aligned with their productivity. We refer to this extension of our baseline model as the "heterogeneous fixed costs" model.

Under the heterogeneous fixed costs model, the emission intensity of a firms given its trading status can be expressed as:

$$\begin{aligned}
\zeta &= \frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}} && \text{for } \mathbb{D},\\
\zeta &= \frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} EP^{\sigma-1}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}^*}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}} && \text{for } \mathbb{M},\\
\zeta &= \frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\left(EP^{\sigma-1}+E^*\left(P^*/\delta\right)^{\sigma-1}\right)\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}} && \text{for } \mathbb{X},\\
\zeta &= \frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\left(EP^{\sigma-1}+E^*\left(P^*/\delta\right)^{\sigma-1}\right)\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\left(\frac{\tilde{\mu}^*}{\phi}\right)^{\frac{1}{1-\rho(\sigma-1)}} && \text{for } \mathbb{Y}.
\end{aligned} \tag{C.2}$$

Using the previous expressions, we can concisely express the natural logarithm of emission intensity as:

$$\begin{aligned}
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) && \text{for } \mathbb{D},\\
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) + \kappa_M && \text{for } \mathbb{M},\\
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) + \kappa_X && \text{for } \mathbb{X},\\
\log(\zeta) &= \kappa - \frac{1}{1-\rho(\sigma-1)}\log(\phi) + \kappa_M + \kappa_X && \text{for } \mathbb{Y},
\end{aligned} \tag{C.3}$$

where the constant term across all types of firms is

$$\kappa \equiv \log\left(\frac{\alpha v_l}{\tau}\left[\left(\frac{\sigma}{\sigma-1}\right)^{-\sigma}\rho\right]^{-\frac{\rho}{1-\rho(\sigma-1)}}\right) - \frac{\rho}{1-\rho(\sigma-1)}\log\left(EP^{\sigma-1}\right) + \frac{1}{1-\rho(\sigma-1)}\log\left(\tilde{\mu}\right),$$

the importer term κ_M is defined as

$$\kappa_M \equiv -\frac{1}{1-\rho(\sigma-1)}\left(\log\left(\tilde{\mu}\right) - \log\left(\tilde{\mu}^*\right)\right),$$

and the exporter term κ_X is defined as

$$\kappa_X \equiv -\frac{\rho}{1-\rho(\sigma-1)}\left\{\log\left[EP^{\sigma-1}+E^*\left(P^*/\delta\right)^{\sigma-1}\right] - \log\left(EP^{\sigma-1}\right)\right\}.$$

Appendix D. Emission intensities normalized by real value-added

Let v denote the real value-added of the firm in our model. As the labor is the only factor of production that creates value-added, the real value-added is defined as $v \equiv wl/p$. Note that the firm's cost minimization problem indicates $wl = pq(1-\alpha)v_l(\sigma-1)/\sigma$. Then it follows that emission intensity normalized by value added, ζ^v, has a linear relation with that normalized by total output, ζ^o, as follows

$$\zeta^v = \frac{zp}{wl} = \frac{\zeta^o}{(1-\alpha)v_l}\frac{\sigma}{\sigma-1}$$

As the coefficients of ζ^o are constants, the comparative statics derived in our main text regarding ζ^o apply immediately to ζ^v.

Table E.1
Balanced test results for the instrument variables.

Panel A							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
VARIABLES	SO_2	COD	NOX	Output	V-Added	SO_2 Rem.	COD Rem.
$\hat{\mathcal{X}}_{ft}$	16.90	−5.721	−4.272	1.035	−1.040	1.961	0.254
	(54.15)	(15.59)	(10.78)	(1.027)	(2.345)	(3.584)	(0.822)
HS6×$D(exp)$ FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Panel B							
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
VARIABLES	NOX Rem.	K / L	K	L	Coal	R&D	SOE
$\hat{\mathcal{X}}_{ft}$	−0.296**	−1156	−1.417	36.57	−3.988	−1.175	−0.00253
	(0.137)	(1 145)	(2.912)	(59.98)	(2.690)	(0.989)	(0.00427)
HS6×$D(exp)$ FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓

Notes:
1. This table presents the regression results for the balance test (Eq. (E.1)) for the validity of our Bartik instruments. The dependent variables in panel A are $\phi_{f,t}$ constructed with firms' total SO_2, COD and NOX emissions, total output, value added, SO_2 and COD removal. The dependent variables in panel B are $\phi_{f,t}$ constructed with firms' total NOX removal, capital-to-labor ratio, total capital, total employment, use of coal, research and development expenditure and state ownership status. 2. All regressions include two-way fixed effects at product-export level and year fixed effects. Standard errors robust against heteroskedasticity and clustered at HS6×$D(exp)$ level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix E. Validation of bartik instrument

Following Borusyak et al. (2022), we confirm that the shift variable in our Bartik IV construction is *as good as randomly assigned*. In what follows, we omit the time subscript t to simplify our illustration. We define $v_h \equiv \sum_f v_{fh}/N_f$ to be the average exposure to product h, in which N_f denotes the total number of firms in a given year and a trade direction, and $\phi_h \equiv \sum_f \frac{v_{fh}\epsilon_f}{N_f v_h}$ is the exposure-weighted average of the residual term ϵ_f in Eq. (20). According to Borusyak et al. (2022), an important identification that justifies the use of Bartik instruments is that the residual term ϵ is orthogonal to the concomitant shift shock. Formally, one needs to show that $\sum_h v_{h,t}\hat{\mathcal{X}}_{h,t}\phi_{h,t} \xrightarrow{P} 0$, where $\hat{\mathcal{X}}_{h,t}$ is the year-to-year trade shock defined as $\hat{\mathcal{X}}_{h,t} \equiv \mathcal{X}_{h,t}/\mathcal{X}_{h,t-1}$.[68]

Intuitively, the validity of the Bartik instrument lies on the assumption that, weighted by v_h, the correlation between product level shocks $\hat{\mathcal{X}}_h$ and the unobservables ϕ_h, which in turn are correlated with firms' initial characteristics, approaches 0 in large sample. Furthermore, the effective sample size is inversely correlated with the Herfindahl–Hirschman index (HHI) of the shares $\mathbb{E}\left(\sum_f v_h^2\right)$, and the IV estimates are consistent when the HHI index is close to zero.[69]

We thus test whether the trade shocks $\hat{\mathcal{X}}_h$ are "balanced" in the sense that they are not correlated with the firm characteristics projected to the product dimension $v_h\phi_h$.

In practice, however, the term ϵ_f is not observed by the researcher. Thus, we follow Campante et al. (2023) and Borusyak et al. (2022) to consider firm characteristics that may be correlated with ϵ_f. These are firms' total SO_2, COD and NOX emissions, total output, value added, removal of SO_2, COD and NOX, capital intensity, capital, employment, R&D and state ownership status. After proxying ϵ in $\phi_{h,t}$ with the aforementioned firm characteristics, we estimate the following equation weighted by $v_{h,t}$:

$$\phi_{h,t} = \beta_0 + \beta_1\hat{\mathcal{X}}_{h,t} + D_{h,t} + \epsilon_{h,t}, \tag{E.1}$$

where $D_{h,t}$ is a set of control variables that include the full set of interaction terms of product and export dummy variables and year fixed effects. If the sign of β_1 is positive, it means that the shocks are positively correlated with unobserved characteristics absorbed by the error term in our main regression Eq. (20).

The regression results presented in Table E.1 assure that all regression coefficients except for that in column 8 are not statistically significant at the conventional level. This means that the shocks at the product level are not systematically correlated with the error term at the firm level. For column (8), the negative sign of the coefficient suggests that product level shocks and firms' NOX removal

[68] To put it differently, one can interpret the condition $\sum_h v_{h,t}\hat{\mathcal{X}}_{h,t}\phi_{h,t} \xrightarrow{P} 0$ as testing the concerns raised by Adao et al. (2019) that shares and the error term may be correlated. Our balance tests around the instrument show that as we use two instruments at the same time, we are unable to use the reg_ss package developed by the authors.

[69] In our sample, we have over 5,000 products and 70,000 firms over approximately 10 years. This suggests that the dimensions of v_{fh} reaches approximately 350 million (= 5000 × 70000), and the averaged export and import shares are close to 0 for the vast majority of the products. As a result, the effective sample size is very large as the HHI index is very low. Bound by our computation capacity, we were not able to calculate the exact HHI index as it entails averaging v_{fh} which has the number of observations equal to the product of firm and product counts (350 million). The problem gets further out of control as our sample spans over 13 years and there are both export and import shares (The number of observations for v_{fh} becomes over 9 billion.). Thus, we are unable to calculate the exact HHI index, and we only use positive v_h in the remainder of analysis in this section.

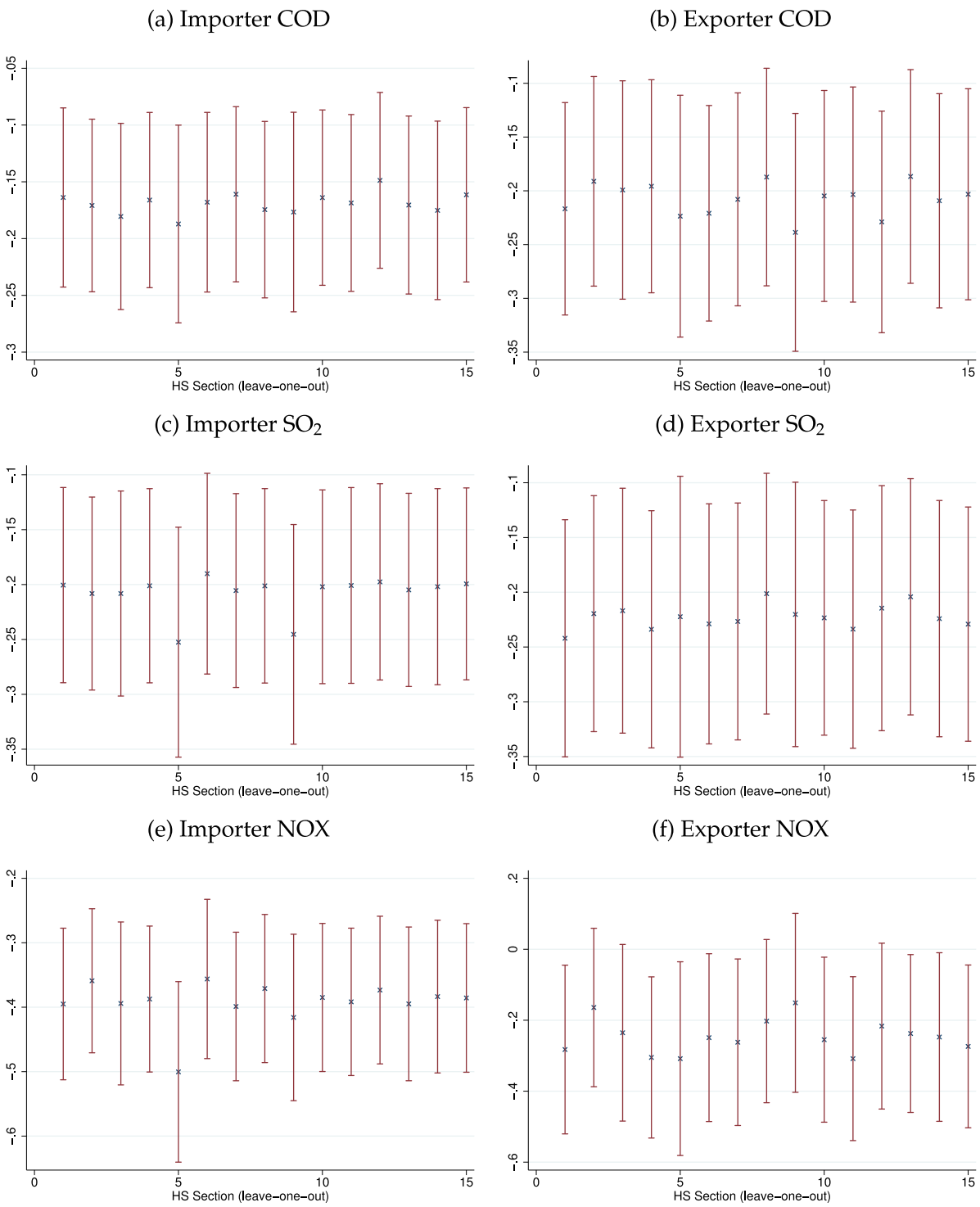

Fig. E.1. Balance test for the Bartik instrument variable. **Notes:** The figures demonstrate the point estimates (crosses in the figures) and the 95% confidence intervals after leaving out one of 15 HS sections to construct the Bartik instruments. The left (right) figures report the estimates for the importer (exporter) status. The regression specifications are the same as columns (1) through (3) in Panel A of Table 5.

are *negatively* correlated. This would happen if firms that mainly export or import a particular product would engage in less NOX removal. This correlation presents an error term structure that is against our hypothesis testing, and thus our coefficients in the main regression are the lower bounds of negative effects of international trade on NOX emission intensity.

Next, we address concerns raised by Goldsmith-Pinkham et al. (2020) that the IV regression results are driven by the correlation between the error term and a subset of products. To examine this, we iteratively construct instrument variables without using shares and shocks for one of 15 sections of HS codes.[70] The idea is to examine the robustness of our results to check whether our IV regression results hinge on a specific subset of products.

We apply this method to our main regression results presented at columns (1) through (3) of panel A of Table 5 and present the point estimates and confidence intervals in Fig. E.1. The figure shows that our point estimates and their inferences are highly robust to leaving out a particular section of HS codes. Minor exceptions appear in the subfigure (f), the exporter coefficient for NOX emission intensities become insignificantly different form zero for 3 of 15 point estimates. Nevertheless, we posit that our instruments are not particularly subject to shocks specific to a particular subset of products and our regression results are not driven by abnormal shocks.

Appendix F. Product level emission intensity

This section describes the procedure to calculate HS 6-digit product-level emission intensities for various emission types.

First of all, we restrict the data samples to the firms that exist in our merged ASIE-CESD-CUST dataset between 2009 and 2013. We choose this time restriction since there might be a technical change in products so that the emission intensity may have changed over a longer period of time. Another benefit of restricting the time to 2009-2013 period is that we can potentially compare our results with those from the previous years.

From the merged ASIE-CESD-CUST dataset, we calculate each exporting product's output share in each firm's total output. We then allocate the firm's total emissions to different HS 6-digit products based on their output shares. We calculate the product level emission intensities after we calculate the total output and emissions associated with each product.

Appendix G. Additional tables and figures

Table G.1
Explanatory power of fixed effects on firm level productivity.

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	$\log(\phi)$ (labor)				$\log(\phi)$ (ACF)	
Observations	481 827	507 767	508 182	482 764	435 164	435 156
R^2	0.734	0.640	0.171	0.284	0.484	0.622
Firm FE	✓	✓			✓	✓
Prov.-Year FE	✓		✓		✓	✓
Ind.-Year FE	✓			✓	✓	✓
Labor productivity						✓

Notes: This table shows the R^2 of the regression results in which the dependent variable is firm-level labor productivity (columns 1 through 4) and the productivity estimated in the style of Ackerberg et al. (2015) (columns 5 and 6).

Table G.2
Domestic vs Importers / Exporters.

	(1)	(2)	(3)	(4)	(5)	(6)
	domestic vs importers			domestic vs exporters		
Emis. Type	COD	SO_2	NOX	COD	SO_2	NOX
κ_M (importer)	−0.148**	−0.105	−0.223***			
	(0.0623)	(0.0658)	(0.0862)			
κ_X (exporter)				−0.231***	−0.147*	−0.0508
				(0.0741)	(0.0768)	(0.166)
Kleibergen-Paap F	441.42	371.80	179.01	303.27	279.74	53.36
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	317238	312818	171394	320187	316425	167687
Firms	58095	55372	39559	58634	56063	38931

Notes:
(i) Columns (1) through (3) in this table compares domestic firms with pure-importing firms (exporting firms are excluded from the sample), whereas columns (4) through (6) in this table compares domestic firms with pure-exporting firms (importing firms are excluded from the sample). (ii) Standard errors robust against heteroskedasticity and clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

[70] There are over 5000 HS 6-digit product codes divided into 15 sections.

Table G.3
Regression Results without Processing Firms.

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	$\log(\zeta^o)$			$\log(\zeta^v)$		
Emis. Type	COD	SO_2	NOX	COD	SO_2	NOX
Panel A. without controlling for productivity						
κ_M (importer)	−0.185***	−0.205***	−0.413***	−0.157***	−0.187***	−0.415***
	(0.0407)	(0.0458)	(0.0620)	(0.0438)	(0.0490)	(0.0699)
κ_X (exporter)	−0.230***	−0.222***	−0.324***	−0.167***	−0.147**	−0.333**
	(0.0509)	(0.0547)	(0.120)	(0.0551)	(0.0588)	(0.135)
Kleibergen-Paap F	475.78	402.81	78.71	475.57	402.72	78.59
p-value ($\kappa_M = \kappa_X$)	0.44	0.78	0.38	0.87	0.56	0.46
Panel B. controlling for productivity						
κ_M (importer)	−0.173***	−0.179***	−0.418***	−0.133***	−0.133***	−0.432***
	(0.0395)	(0.0446)	(0.0604)	(0.0390)	(0.0439)	(0.0614)
κ_X (exporter)	−0.226***	−0.213***	−0.324***	−0.160***	−0.130**	−0.336***
	(0.0494)	(0.0533)	(0.118)	(0.0490)	(0.0529)	(0.120)
β_ϕ (productivity)	−0.346***	−0.342***	−0.255***	−0.706***	−0.705***	−0.623***
	(0.00395)	(0.00410)	(0.00489)	(0.00417)	(0.00434)	(0.00564)
Kleibergen-Paap F	475.69	402.68	78.71	475.48	402.59	78.59
p-value ($\kappa_M = \kappa_X$)	0.35	0.57	0.34	0.63	0.96	0.33
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	407538	383936	211573	407630	384028	211669
Firms	68858	64422	46794	68863	64423	46798

Notes:
(i) This table shows the IV regression results of firm-level emission intensities on importer and exporter statuses. The regression specification is presented in Eq. (20). ζ^o and ζ^v denote the firm level emission intensities normalized by the value of real total output and value-added, respectively. κ_M and κ_X are the coefficients of the dummy variables for the importer and exporter statuses respectively. β_ϕ is the coefficient of the natural logarithm of labor productivity. p-value reports the result of the t-test where the null hypothesis assumes $\kappa_M = \kappa_X$. The instruments for the endogenous variables are discussed in Section 3.4.
(ii) Panels A and B show the regression results without and with firm-level productivity in the regressors.
(iii) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

Table G.4
Regression Results with ACF Productivity.

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	$\log(\zeta^o)$			$\log(\zeta^v)$		
VARIABLES	COD	SO_2	NOX	COD	SO_2	NOX
κ_M (importer)	−0.176***	−0.214***	−0.339***	−0.162***	−0.201***	−0.342***
	(0.0416)	(0.0482)	(0.0590)	(0.0428)	(0.0493)	(0.0621)
κ_X (exporter)	−0.215***	−0.241***	−0.154	−0.186***	−0.211***	−0.197
	(0.0533)	(0.0581)	(0.120)	(0.0551)	(0.0597)	(0.128)
β_ϕ (ACF productivity)	−0.172***	−0.192***	−0.0256**	-1.169***	-1.193***	-1.034***
	(0.00972)	(0.00996)	(0.0104)	(0.0217)	(0.0225)	(0.0208)
Kleibergen-Paap F	438.68	365.53	73.77	438.68	365.53	73.77
p-value ($\kappa_M = \kappa_X$)	0.52	0.69	0.08	0.71	0.88	0.19
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	381956	358554	191316	381956	358554	191316
Firms	64575	60733	43134	64575	60733	43134

Notes:
(i) This table shows the IV regression results of firm-level emission intensities on importer and exporter statuses. The regression specification is presented in Eq. (20). ζ^o and ζ^v denote the firm level emission intensities normalized by the value of real total output and value-added, respectively. κ_M and κ_X are the coefficients of the dummy variables for the importer and exporter statuses respectively. β_ϕ is the coefficient of the natural logarithm of the ACF productivity. p-value reports the result of the t-test where the null hypothesis assumes $\kappa_M = \kappa_X$. The instruments for the endogenous variables are discussed in Section 3.4. ACF productivity denotes the productivity level estimated à la Ackerberg et al. (2015).
(ii) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

Table G.5
The role of product mix.

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	$\log(\zeta^o)$			$\log(\zeta^v)$		
Emis. Type	COD	SO_2	NOX	COD	SO_2	NOX
Panel A. domestic firms vs. firms with 4 or fewer export products						
κ_M (importer)	−0.178***	−0.203***	−0.358***	−0.130***	−0.157***	−0.282***
	(0.0428)	(0.0461)	(0.0779)	(0.0458)	(0.0494)	(0.0859)
κ_X (exporter)	−0.263***	−0.215***	−0.482***	−0.201***	−0.124*	−0.389*
	(0.0636)	(0.0665)	(0.185)	(0.0684)	(0.0716)	(0.204)
Kleibergen-Paap F	316.12	294.53	31.37	316.06	294.48	31.31
p-value ($\kappa_M = \kappa_X$)	0.22	0.87	0.36	0.33	0.67	0.47
Obs.	374820	361359	195309	374911	361451	195405
Firms	65556	61903	43966	65561	61904	43970
Panel B. domestic firms vs. firms with more than 4 export products						
κ_M (importer)	−0.118**	−0.154***	−0.428***	−0.0812	−0.139**	−0.439***
	(0.0504)	(0.0571)	(0.0734)	(0.0544)	(0.0608)	(0.0821)
κ_X (exporter)	−0.296***	−0.308***	−0.226	−0.215**	−0.220**	−0.268
	(0.0843)	(0.0949)	(0.195)	(0.0919)	(0.102)	(0.216)
Kleibergen-Paap F	259.75	214.68	58.01	259.70	214.68	58.01
p-value ($\kappa_M = \kappa_X$)	0.07	0.15	0.34	0.21	0.49	0.47
Firm FE	✓	✓	✓	✓	✓	✓
Prov.-Year FE	✓	✓	✓	✓	✓	✓
Ind.-Year FE	✓	✓	✓	✓	✓	✓
Obs.	365493	348953	191765	365578	349041	191857
Firms	65177	61233	44080	65182	61234	44084

Notes:
(i) This table shows the IV regression results of firm-level emission intensities on importer and exporter statuses. The regression specification is presented in Eq. (20). ζ^o and ζ^v denote the firm level emission intensities normalized by the value of real total output and value-added, respectively. κ_M and κ_X are the coefficients of the dummy variables for the importer and exporter statuses respectively. p-value reports the result of the t-test where the null hypothesis assumes $\kappa_M = \kappa_X$. The instruments for the endogenous variables are discussed in Section 3.4.
(ii) Panel A (B) compares domestic firms with trading firms that export 4 or fewer (more than 4) products.
(iii) Standard errors robust against heteroskedasticity and clustered at the firm level are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

Table G.6
List of top emission intensive products.

HS 6-digit Code	Product Description
Panel A. Top COD-intensive Products	
470411	Wood pulp of coniferous wood
470319	Wood pulp of non-coniferous wood
380400	Lyes from the manufacture of wood pulp
293626	Vitamin B12 and its derivatives
310520	Fertilizers, mineral or chemical (nitrogen, phosphorus and potassium)
Panel B. Top SO_2-intensive Products	
540810	Fabrics, woven; of viscose rayon
470411	Wood pulp of non-coniferous wood
310280	Fertilizers, mineral or chemical (urea and ammonium nitrate)
521151	Fabrics, woven; containing less than 85% by weight of cotton
790310	Zinc dust
Panel C. Top NOX-intensive Products	
470411	Wood pulp of coniferous wood
310280	Fertilizers, mineral or chemical (urea and ammonium nitrate)
252310	Cement clinkers
540810	Fabrics, woven; of viscose rayon
700312	Cast glass and rolled glass

Notes: This table shows the lists of top 5 most emission-intensive HS 6-digit product codes by different types of emissions. Source: ASIE, CESD, Customs Data and authors' calculations.

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